

AI Discourse at Scale:  
Topic, Sentiment, and Community Dynamics Across Reddit

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# 1. Introduction

## 1.1 Project Motivation

In 2025, it's impossible to move through daily life without hearing the word AI. Artificial intelligence has woven itself into everything—from homework help to healthcare, from political conversations to the memes circulating at 2 a.m. Statistics from the past two years show unprecedented adoption curves: millions of new users interacting with AI tools monthly, entire industries restructuring around automation, and social platforms filled with debates about what AI means for work, creativity, and the future.

Beyond the fact that our team met in a Big Data and Cloud Computing Course, we shared a deeper curiosity: *what did earlier, rawer conversations about AI actually look like?* During the hype, as corporate messaging took over, as every tech company had a chatbot on its homepage, how were people processing this technological shift?

Reddit offered a natural window into that moment. Unlike platforms optimized for virality, Reddit tends to produce longer, more thoughtful and unfiltered threads, where users can argue, joke, panic, speculate, analyze, and sometimes spiral—often with more intellectual honesty than faster-paced platforms like Twitter.

Together, we converged on this question:

**With the advent of AI, how were online communities processing that shift in real time? Who was talking? What were they saying? And how did different AI subreddits evolve as the technology accelerated?**

This project is our attempt to answer those questions by examining discussions from 2023–2024—the peak of the AI boom—using large-scale EDA, NLP, and machine learning. In many ways, studying these conversations offers insights not only into AI discourse, but into how societies could have historically reacted to revolutionary technologies, more broadly.

## 1.2 Overview of Subreddits Studied

Our dataset consists of posts and comments drawn from 15 AI- and technology-related subreddits active from **June 1, 2023** to **July 31, 2024**. These communities were selected because they represent a broad spectrum of AI discourse, ranging from casual user discussions to technical research to societal debates about the future of automation.

| Thematic Category                                         | Subreddits                                                                                                                        | Purpose                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Discussion &amp; Public Sentiment</b>          | <a href="#">AskReddit</a>                                                                                                         | Our <b>benchmark</b> subreddit. All types of questions, of any kind, are posted here. As of today, AskReddit hosts over <b>17M weekly users</b> . Because of its scale, AskReddit is particularly effective for capturing large-scale public reactions to AI whenever an AI-related question is posted. Threads here often go viral and reflect broad, opinion-driven engagement from non-expert users. |
| <b>AI Product &amp; Model Communities</b>                 | <a href="#">ChatGPT</a> , <a href="#">ClaudeAI</a> , <a href="#">PerplexityAI</a> , <a href="#">OpenAI</a> , <a href="#">GPT4</a> | Centers on direct user interaction with AI products, such as chatbots. Discussions include troubleshooting, model updates, use cases, and early reactions to new capabilities.                                                                                                                                                                                                                          |
| <b>Open-Source Model Development &amp; Creative Tools</b> | <a href="#">LocalLLaMA</a> , <a href="#">StableDiffusion</a>                                                                      | Focuses on hands-on experimentation with open-source AI tools. Users fine-tune models, share prompts, test releases, and innovate outside major corporate ecosystems.                                                                                                                                                                                                                                   |
| <b>Technical, Research, and Employment Communities</b>    | <a href="#">MachineLearning</a> , <a href="#">datascience</a> , <a href="#">programming</a> , <a href="#">computerscience</a>     | Represents expert and practical discussions around ML theory, benchmarks, implementations, and real-world applications.                                                                                                                                                                                                                                                                                 |
| <b>Societal, Political, and Futurist Perspectives</b>     | <a href="#">Futurology</a> , <a href="#">neoliberal</a> , <a href="#">singularity</a>                                             | Engages with AI's broader impacts: policy debates, economic implications, long-term societal trajectories.                                                                                                                                                                                                                                                                                              |

### 1.3 Dataset Source

This dataset was drawn from a larger Reddit corpus consisting of multiple terabytes of data and tens of billions of total rows. Using Spark, we filtered the full dataset to include only the 15 AI-related subreddits described in Section 1.2. The resulting AI-centric dataset contains approximately 8 billion rows, including both comments and submissions. All data was provided in Parquet format, which allowed for efficient distributed processing across a Spark cluster.

### 1.3.1. Scale and Schema

The dataset includes two schemas—one for comments and one for submissions. Each contains user information, timestamps, text content, and engagement metrics. Coverage across subreddits is consistent for the entire 14-month period, enabling longitudinal comparisons of engagement, sentiment, and posting volume.

#### 1.3.1.1. Submissions Schema

| Field        | Datatype   | Description                                                             |
|--------------|------------|-------------------------------------------------------------------------|
| id           | string     | Unique identifier for the submission (post).                            |
| subreddit    | string     | Community where the post was published.                                 |
| author       | string     | Username of the poster (may be "[deleted]").                            |
| title        | string     | Title of the submission. Often the main content for link posts.         |
| selftext     | string     | Body text of the post (empty for link/media posts; may be "[removed]"). |
| score        | int        | Net upvotes (upvotes – downvotes).                                      |
| created_utc  | int (Unix) | Time the submission was created.                                        |
| num_comments | int        | Total number of comments on the submission.                             |
| url          | string     | Link associated with the post (external or internal).                   |
| over_18      | boolean    | NSFW flag; true if the post is marked as adult content.                 |
| date         | string     | Derived posting date. (YYYY-MM-DD)                                      |

#### 1.3.1.2. Comment Schema

| Field     | Datatype | Description                                         |
|-----------|----------|-----------------------------------------------------|
| id        | string   | Unique identifier for the comment.                  |
| subreddit | string   | Name of the subreddit where the comment was posted. |
| author    | string   | Username of the commenter (may be "[deleted]").     |



|                  |            |                                                                               |
|------------------|------------|-------------------------------------------------------------------------------|
| body             | string     | Text content of the comment.                                                  |
| score            | int        | Net upvotes (upvotes – downvotes).                                            |
| created_utc      | int (Unix) | Time the comment was created (in seconds since epoch).                        |
| parent_id        | string     | ID of the parent comment or submission (t1_ for comment, t3_ for submission). |
| link_id          | string     | ID of the submission the comment belongs to.                                  |
| controversiality | int        | Indicator of vote disagreement (0 = normal, 1 = controversial).               |
| gilded           | int        | Number of awards (“golds”) the comment received.                              |
| date             | string     | Derived date field (YYYY-MM-DD)                                               |

Fields highlighted in **green** represent the variables we thought were most relevant to our research questions, including subreddit, author, body/selftext, score, num\_comments, created\_utc, and structural identifiers such as parent\_id and link\_id.

### 1.3.2. Considerations

A defining feature of this dataset is its extreme imbalance across subreddits.

Community sizes vary widely:

- AskReddit contains over **58 million rows** and dominates the dataset.
- Mid-sized AI communities (e.g., ChatGPT, LocalLLaMA, StableDiffusion) contain hundreds of thousands to a few million rows.
- Emerging communities (GPT4, PerplexityAI) contain only several thousand row

| Subreddit   | # Comments | # Submissions | Row        | Avg Score |
|-------------|------------|---------------|------------|-----------|
| AskReddit   | 55,851,868 | 2,686,082     | 58,537,950 | 10.12     |
| neoliberal  | 4,673,171  | 35,352        | 4,708,523  | 38.27     |
| ChatGPT     | 1,748,893  | 145,862       | 1,894,755  | 28.91     |
| singularity | 1,166,552  | 37,112        | 1,203,664  | 24.85     |
| Futurology  | 898,463    | 15,884        | 914,347    | 91.45     |

|                 |         |        |         |       |
|-----------------|---------|--------|---------|-------|
| StableDiffusion | 763,492 | 74,526 | 838,018 | 12.37 |
| LocalLLaMA      | 425,095 | 31,020 | 456,115 | 11.93 |
| OpenAI          | 337,768 | 25,574 | 363,342 | 14.36 |
| programming     | 325,140 | 31,396 | 356,536 | 14.12 |
| datascience     | 193,664 | 24,717 | 218,381 | 5.23  |
| MachineLearning | 146,901 | 37,191 | 184,092 | 4.31  |
| ClaudeAI        | 59,790  | 5,097  | 64,887  | 6.87  |
| computerscience | 31,181  | 7,504  | 38,685  | 3.81  |
| GPT4            | 4,009   | 1,623  | 5,632   | 1.65  |
| PerplexityAI    | 33      | 28     | 61      | 1.19  |

This imbalance has major implications for exploratory analysis and modeling:

- Aggregate statistics can be dominated by large subreddits unless stratified.
- Engagement patterns differ substantially by community (e.g., a score of 10 is average in *AskReddit* but high in *ChatGPT* and extremely low in *Futurology*).
- Smaller subreddits lack the statistical power for some analyses and require sampling or qualitative interpretation.
- Machine learning models must account for imbalance using techniques like stratified sampling, oversampling smaller communities, or subreddit-specific modeling.

Because of these considerations, all EDA and ML tasks in this project are performed per subreddit or with explicit normalization/stratification. This ensures that the *unit of analysis remains meaningful* and that insights reflect genuine community behavior.

## 2. Data Cleaning & Preparation

### 2.1 Duplicate and Missing Identifiers Checks

We first examined **duplicate IDs** for submissions and comments.

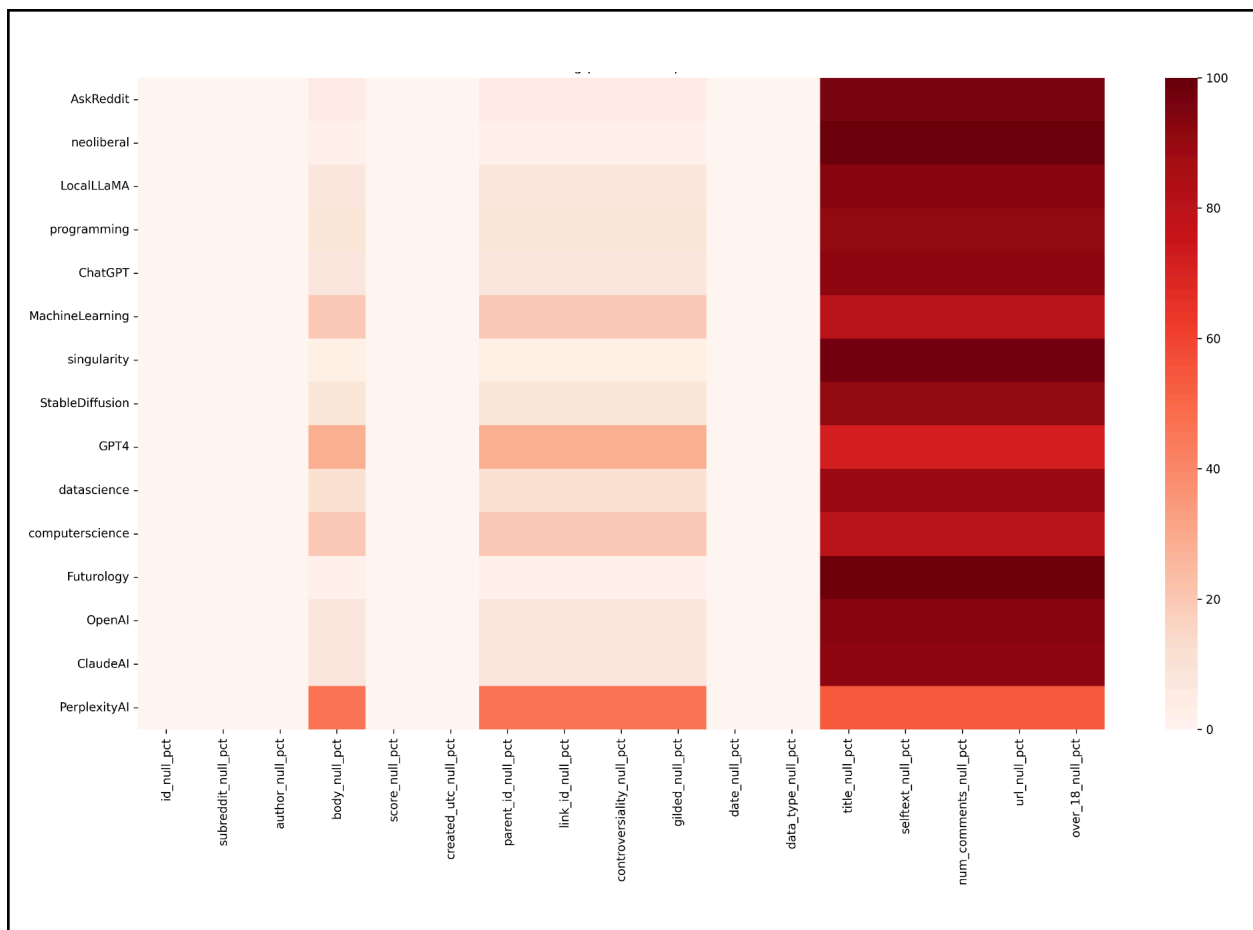
- No duplicate post or comment IDs were found at the subreddit level.

- This indicates that the dataset does not contain accidental re-ingestion or redundant snapshots.

We also confirmed that all rows include a valid `id` field. The only potential duplicates will be `user_ids`, which will be examined during EDA.

## 2.2 Missingness & Null Values Per Subreddits

We profiled missingness per column per subreddit to assess structural gaps in the dataset.



### 2.2.1 Key Observations:

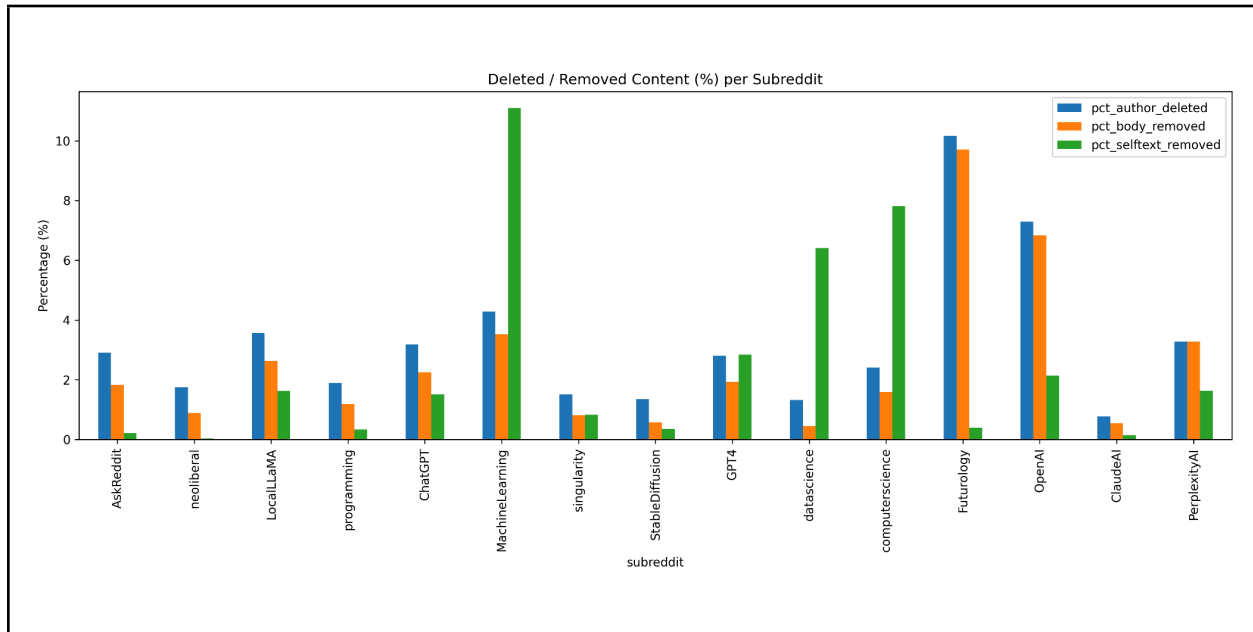
Structural fields like `id`, `subreddit`, `author`, `score`, `created_utc`, `date`, and `datatype` have no missing values across communities. Posts without a `body`, `selftext`, or `title` correspond exactly to comment-submission schema differences, not quality issues. Some subs (e.g., AskReddit) show large “missing” counts for submission-only fields, simply because they have far more comments than posts.

### 2.2.2 Conclusion:

Most null values arise from the schema, not data corruption. This means no imputation or row removal was necessary.

## 2.3 Deleted & Removed Content

Reddit often masks content using [deleted] (for authors) or [removed] (for text). We quantified the percent of content removed per subreddit. These values matter because downstream NLP tasks—topic modeling or sentiment—could be skewed by many empty text fields.



### 2.3.1 Key Observations:

Structural fields like id, subreddit, author, score, created\_utc, date, and datatype have no missing values across communities. Posts without a body, selftext, or title correspond exactly to comment-submission schema differences, not quality issues. Some subs (e.g., AskReddit) show large “missing” counts for submission-only fields, simply because they have far more comments than posts.

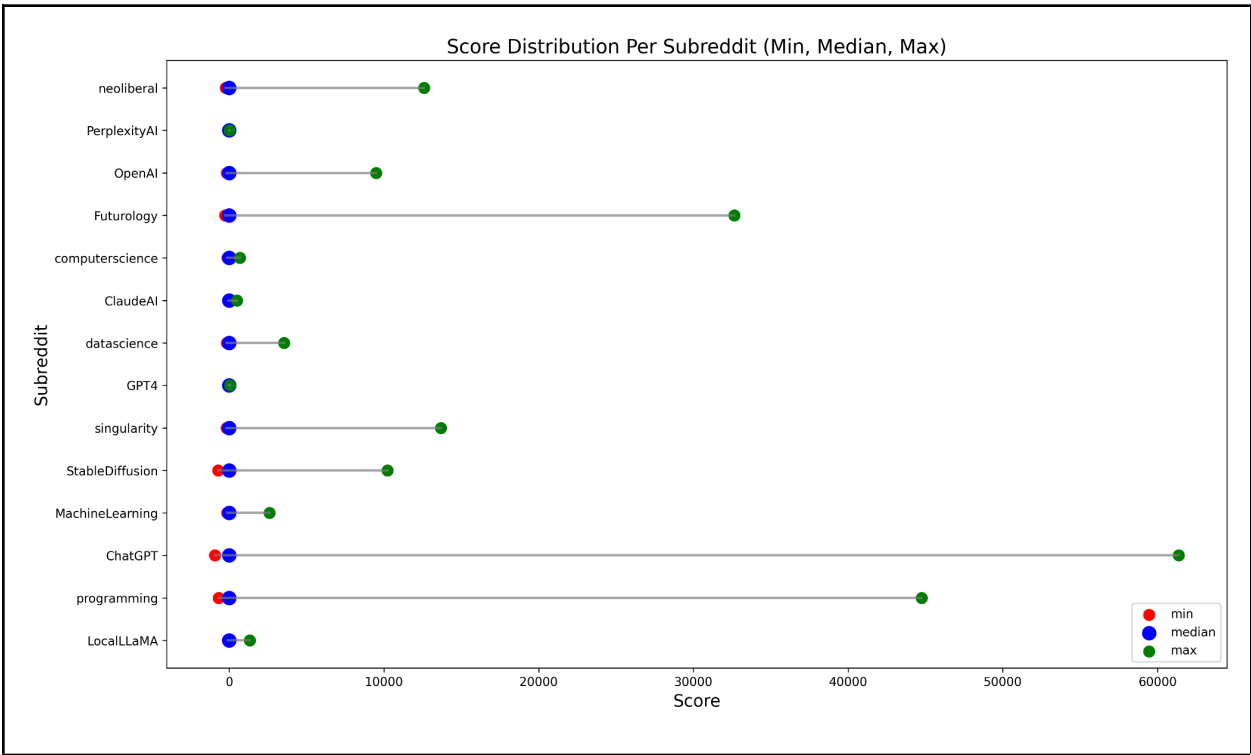
### 2.3.2 Conclusion:

Most null values arise from the schema, not data corruption. This means no imputation or row removal was necessary.

## 2.4 Score Distribution & Outlier Analysis

### 2.4.1 Score Distribution per Subreddit (Min–Median–Max)

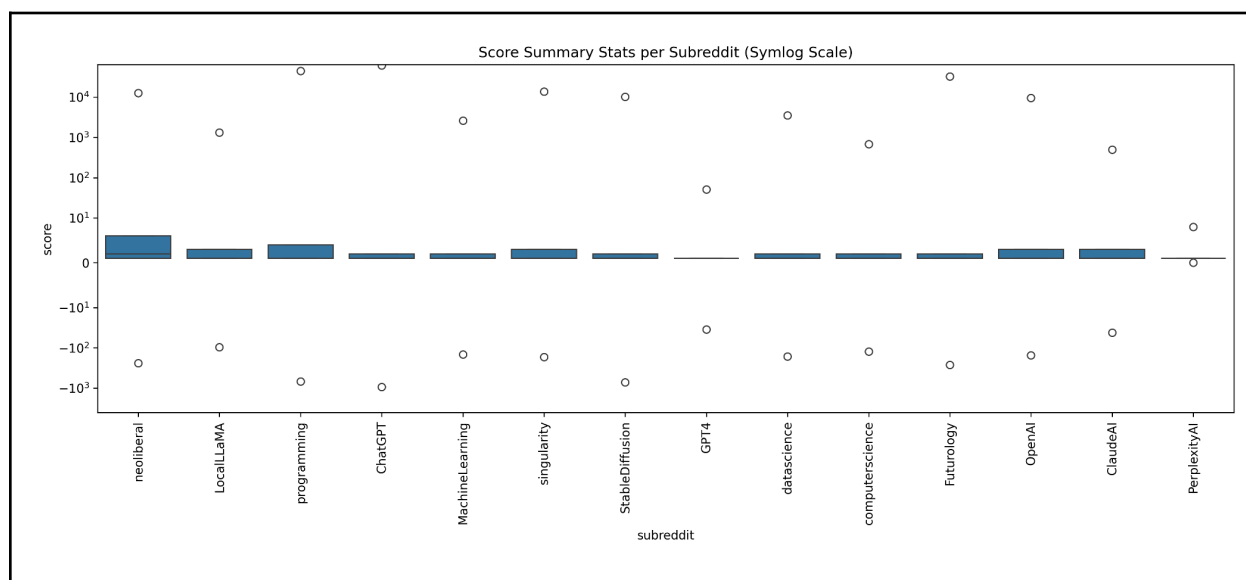
The score distribution across AI-related subreddits shows a pronounced long-tailed pattern, where most comments receive little to no engagement while a small number of posts achieve disproportionately high visibility. In nearly all subreddits, the median score remains extremely low—often near zero—indicating that the majority of contributions garner minimal attention. However, the maximum values reveal substantial variation in a subreddit’s potential for viral content. Communities such as ChatGPT, programming, Futurology, and PerplexityAI show exceptionally high maximum scores, frequently reaching tens of thousands. This suggests that posts featuring practical coding tips, demonstrations of AI capabilities, or discussions about the societal implications of future technologies tend to generate widespread interest. The neoliberal subreddit also displays comparatively elevated maximum scores, implying that AI content framed within political or economic contexts can likewise provoke significant engagement. Overall, the min–median–max distribution highlights that while participation is broad, high-impact discussions are concentrated within specific communities and topical themes.



### 2.4.2 Score Summary Stats per Subreddit (Log-Scale Boxplot)

The log-scaled score summary plot further reinforces the highly skewed nature of engagement across AI-related subreddits. Even after transformation, most subreddits exhibit compressed interquartile ranges with a dense cluster of low-score values, while producing numerous high-score outliers that extend far upward on the logarithmic scale. These outliers represent the

small fraction of comments that capture disproportionate attention and drive the observable long-tail dynamic. Subreddits such as ChatGPT and programming show the most extensive upper tails, confirming their tendency to generate frequent high-visibility posts. Similarly, Futurology and neoliberal exhibit significant upward spread, suggesting that conversations about the future of AI, geopolitical implications, or political economy resonate strongly with users and often lead to elevated engagement. Taken together, the log-scale distribution illustrates that AI discourse on Reddit is simultaneously widespread and uneven: while most comments remain low-visibility, select topics—particularly those tied to technological practice, future speculation, or political framing—consistently produce highly amplified reactions.



## 3. Exploratory Data Analysis (EDA)

### 3.1 AI Subreddit Engagement Analysis

#### 3.1.1 Business Question

How do AI subreddit categories differ in user engagement, and how do title characteristics — such as length and question phrasing—shape the number of comments and scores that posts receive?

#### 3.1.2 Analysis Approach

Specific subreddits were mapped into six broader AI community categories to enable comparative analysis across different discourse types:

- **Chatbot-Focused:** ChatGPT, OpenAI, GPT4, ClaudeAI, PerplexityAI

- **Creative-Generation:** StableDiffusion, MidJourney, Sora, AI Art
- **Research/Technical:** Generative AI, Artificial Intelligence, Machine Learning, computer science, data science, programming
- **Social/Policy:** Futurology, singularity, neoliberal
- **Dev/Hardcore:** Local LLaMA, OpenAI\_Dev
- **Baseline Control:** AskReddit

This categorization allows us to identify patterns specific to different types of AI discourse while maintaining sufficient sample sizes for robust statistical analysis.

### 3.1.3 Objective 1: Identify Highest Engagement Categories

#### Methodology

Aggregated per-category engagement metrics including:

- Number of posts
- Average title length and word count
- Frequencies of question patterns (question\_pct, how\_pct, what\_pct, why\_pct)
- Detailed engagement statistics including mean, median, and maximum comments and scores
- Standard deviations and percentile approximations

Expected output: summary\_statistics.csv file containing comprehensive category-level metrics.

#### Results

Social policy shows the highest average of 59 comments and the highest score potential, with extremely viral posts (max score of 32,643). Users in this segment respond strongly to short, impactful titles and to emotionally or socially charged topics.

#### Summary Statistics by Category:

| Category            | Total Posts | Avg Title Length | Question % | Avg Comments | Median Comments | Avg Score | Max Score |
|---------------------|-------------|------------------|------------|--------------|-----------------|-----------|-----------|
| Social/Policy       | 88,348      | 71.4             | 18.5%      | 59.2         | 1               | 78.7      | 32,643    |
| Baseline Control    | 2,686,082   | 76.5             | 95.4%      | 15.4         | 1               | 9.6       | 77,955    |
| Dev/Hardcore        | 31,020      | 58.1             | 36.6%      | 10.4         | 1               | 19.6      | 1,327     |
| Chatbot-Focused     | 178,184     | 54.8             | 25.3%      | 8.1          | 1               | 45.8      | 61,374    |
| Creative-Generation | 74,526      | 54.2             | 25.9%      | 7.0          | 1               | 21.1      | 10,227    |
| Research/Technical  | 100,808     | 57.3             | 24.4%      | 4.9          | 1               | 9.8       | 44,747    |

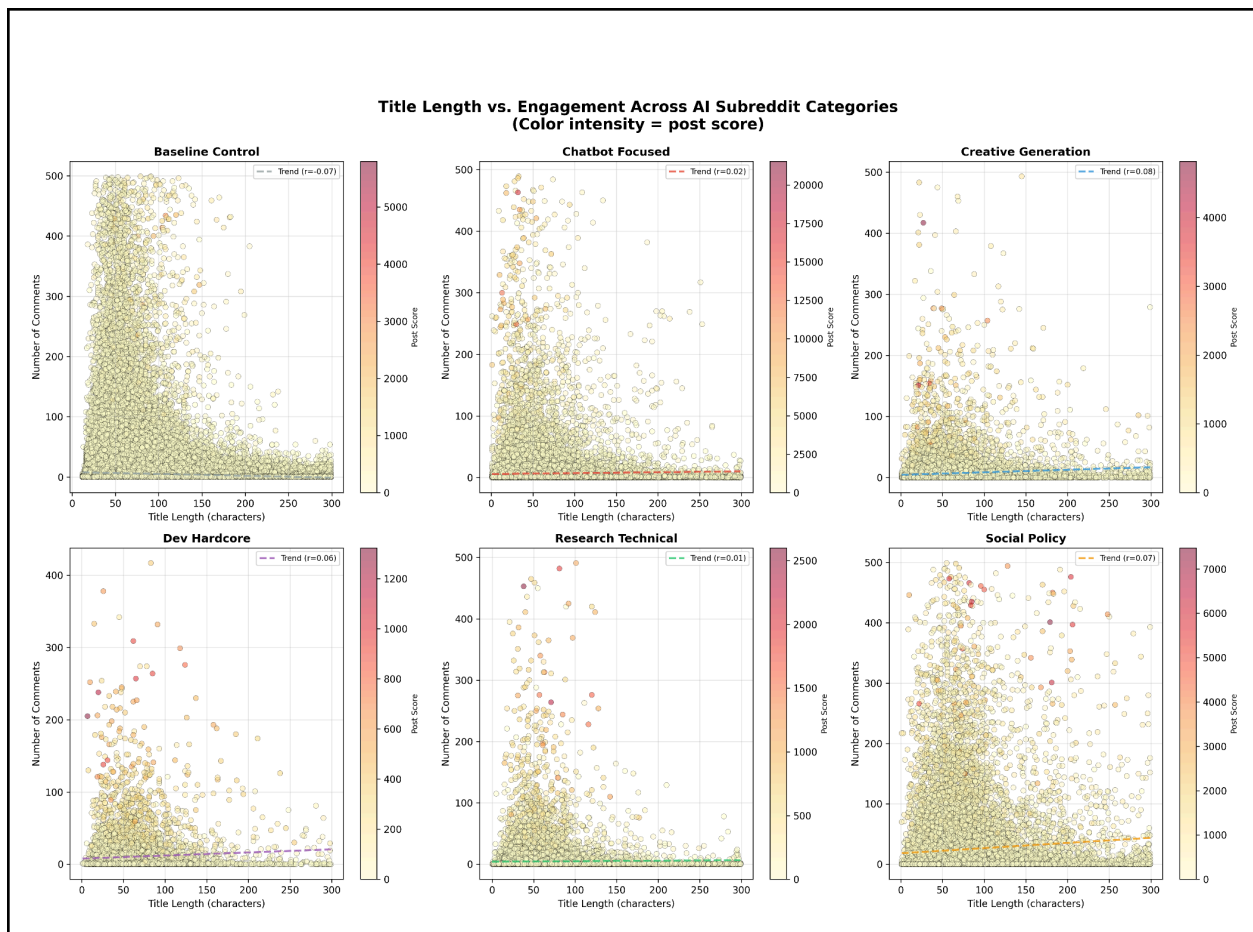
### 3.1.4 Objective 2: Title Length Correlation with Engagement

#### Methodology

Analysis approach:

- Filtered extreme outliers where comments are less than 500 and title length is less than 300
- Used a 30% sample with a fixed seed of 42 for reproducibility
- Generated scatter plots with trend lines and color-encoded scores

Expected output: Scatter plot visualizations showing title length vs. engagement metrics across all categories.



#### Results

Title length shows a weak direct correlation with overall engagement. The Pearson correlation coefficient is near zero (-0.07 to 0.08) across all categories. The engagement is heavily concentrated in shorter titles, which are less than 100 characters, across nearly all categories.

This finding challenges conventional assumptions that longer, more detailed titles generate more engagement. Instead, brevity appears to be valued across all AI community types, with



the most successful posts using concise, impactful phrasing regardless of whether they're technical discussions or social debates.

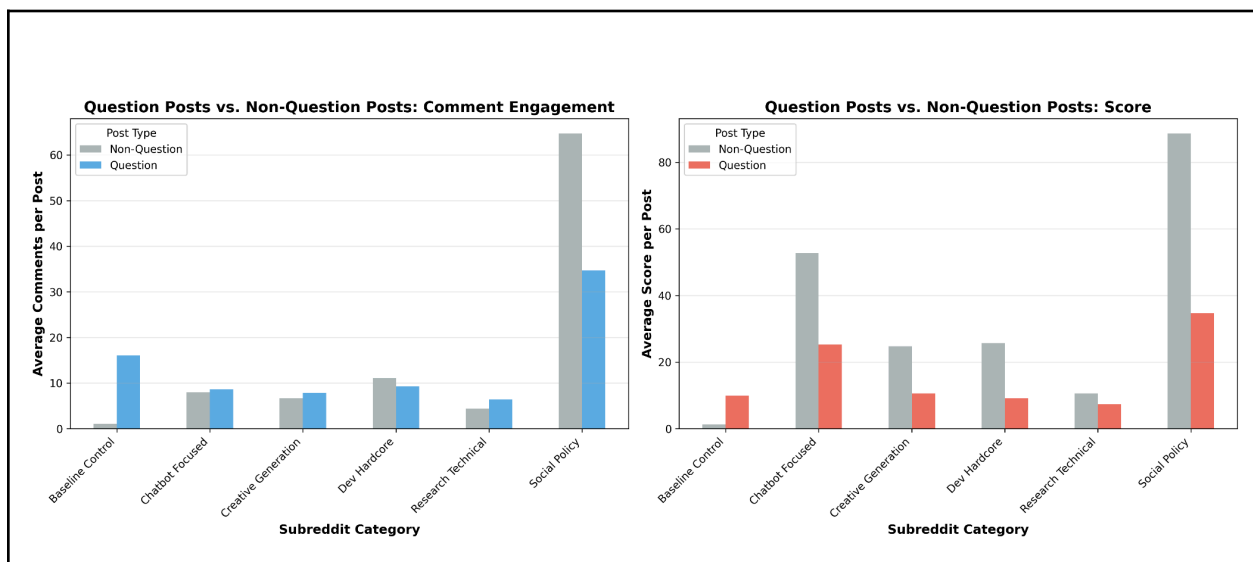
### 3.1.5 Objective 3: Question vs. Non-Question Title Engagement

#### Methodology

Analysis approach:

- Compared engagement for question-style vs. statement-style titles across each AI subreddit category
- Grouped data by is\_question binary feature
- Computed number of posts, average comments, and average scores for each group
- Pivoted results and visualized as side-by-side bar charts

Expected output: Side-by-side bar charts of average comments and average score grouped by AI subreddit category and colored by post type (Question vs. Non-Question).



#### Results

Posts phrased as questions do not universally outperform non-questions. The baseline control group shows much higher engagement with question-styled posts, while social/policy Reddit communities strongly favor non-question statements across both comment engagement and scores.

In analyzing the data, it is evident that most technical and creative categories generated a higher number of questions relative to average comment engagement. Conversely, posts comprised of statements demonstrated significantly higher average scores. This pattern suggests that different community types have distinct norms around what constitutes engaging content—technical communities value inquiries that prompt detailed explanations, while debate-oriented communities respond to provocative statements that spark discussion.

## 3.2 Engagement Analysis

### 3.2.1 Business Question

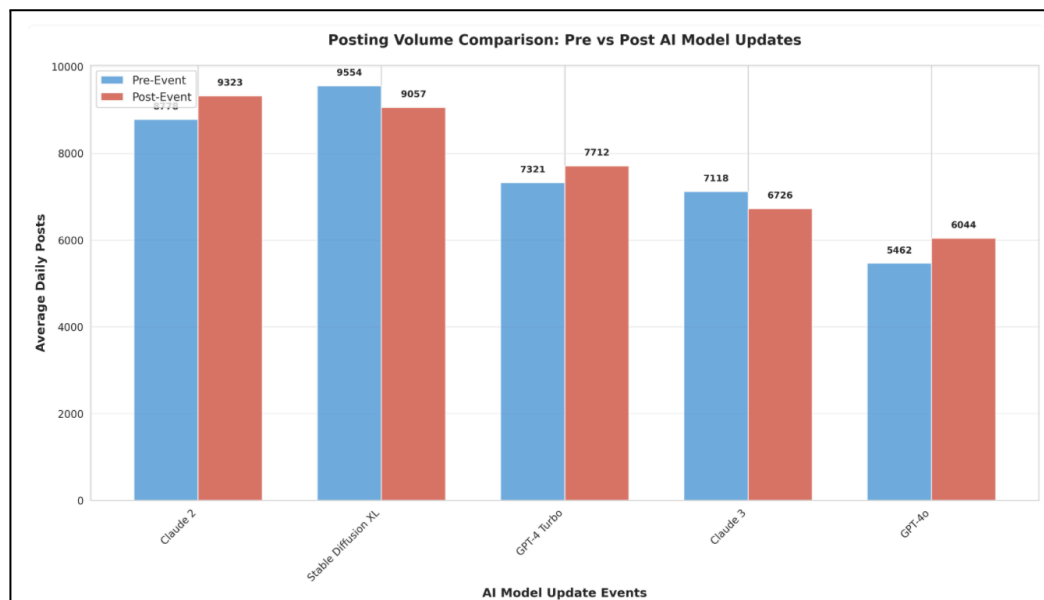
How do engagement levels change before and after major AI model updates, and which subreddits exhibit the strongest and most persistent hype spikes?

### 3.3.2 Analysis Approach

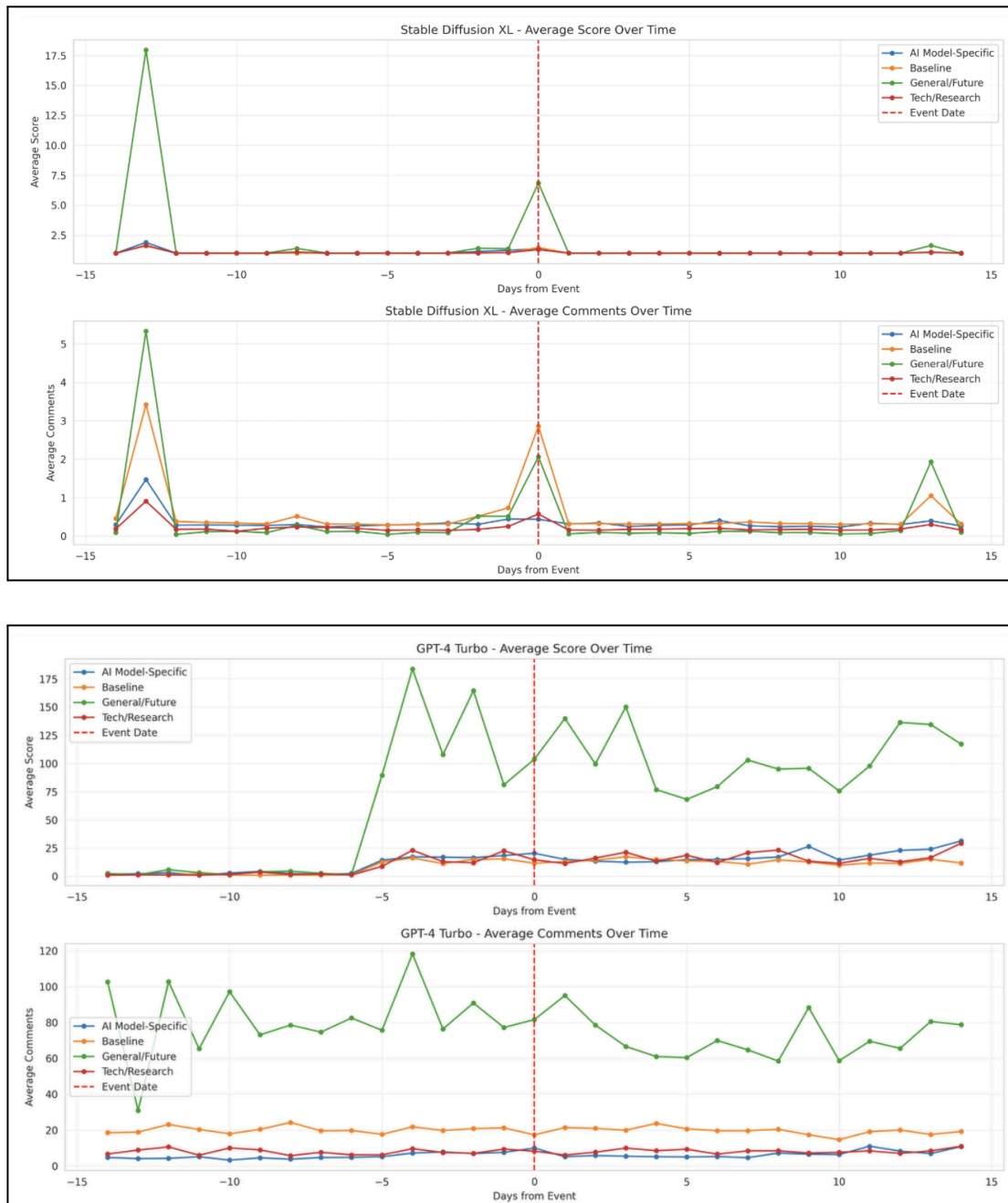
- Five major model updates (Claude 2, GPT-4 Turbo, SDXL, Claude 3, GPT-4o) served as anchor events. We analyzed a **28-day window** for each event, comparing the **Pre-Event** period (14 days prior) against the **Post-Event** period (14 days after) to isolate immediate reactions from long-term trends.
- Daily averages for **Post Volume**, **Sentiment Score**, and **Comment Count** were aggregated. Crucially, subreddits were stratified into **Model-Specific** (e.g., r/ChatGPT), **General Tech**, and **Baseline** categories to distinguish between niche enthusiasm and broader public interest.
- **Impact Classification:** By comparing pre- and post-event means, we classified community reactions into distinct behavioral patterns: *Short-Term Hype Spikes*, *Sustained Growth*, or *Negative/Neutral Contraction*.

### 3.2.3 Findings: Impact of Model Releases on Engagement

Our analysis of major AI model releases (Claude 2, GPT-4 Turbo, Stable Diffusion XL, Claude 3, GPT-4o) reveals distinct patterns in community engagement. While hype spikes are a common phenomenon, they are neither universal nor uniform across different communities.



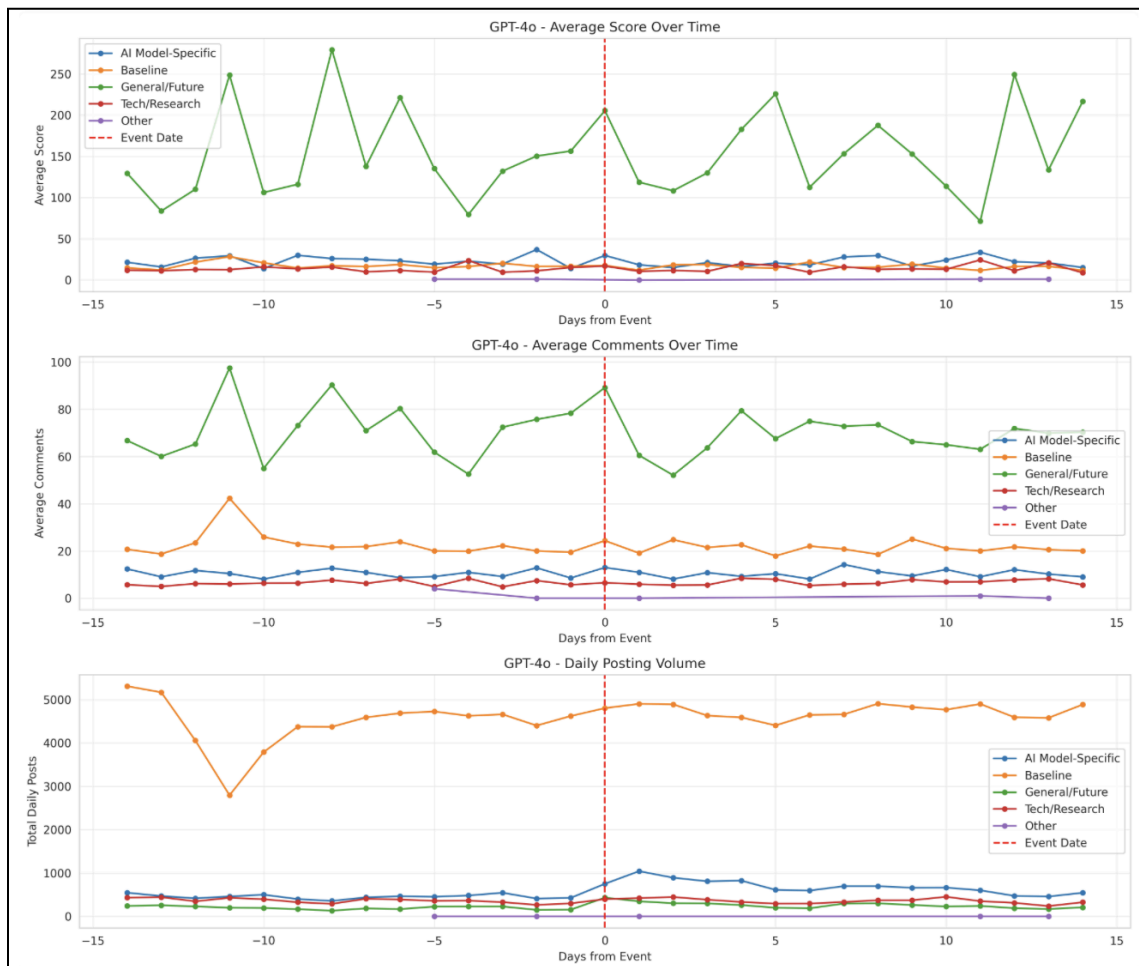
**1. Immediate "Hype Spikes" are Prevalent but Short-Lived** Most major model releases trigger an immediate reaction in engagement metrics, specifically in average scores and comment counts.



- **Rapid Decays:** As seen with **Stable Diffusion XL**, engagement can spike dramatically on the event date (Day 0) across "Baseline" and "General/Future" communities, only to return to pre-event levels within 2-3 days.

- **Anticipatory Activity:** In some cases, such as **GPT-4 Turbo**, significant spikes in average scores occur in "General/Future" subreddits *prior* to the official release (Day -5), suggesting high levels of community speculation and anticipation.

**2. Uneven Impact on Discussion Volume** Contrary to the expectation that every major release drives traffic, the impact on daily posts volume varies significantly by model success and community reception.





- **Growth Events: GPT-4o** demonstrated a clear successful release, driving a visible increase in daily post volume from a pre-event average of 5,462 to a post-event average of 6,044. Similarly, **Claude 2** saw a volume increase from 8,778 to 9,323 daily posts.
- **Contraction Events:** Conversely, releases like **Claude 3** and **Stable Diffusion XL** actually saw slight *decreases* in average daily posts in the two weeks following the event (Claude 3 dropped from 7,118 to 6,726). This suggests that without a perceived "technological leap," new releases fail to sustain broader community attention.

**3. Divergence Between Specialized and General Communities** The data highlights a functional split in how different subreddits react to news.

- **Model-Specific Consistency:** Dedicated subreddits (the "AI Model-Specific" category, represented by blue lines) tend to show more stable, sustained increases in posting volume following releases. For example, following the **GPT-4o** release, the model-specific posting volume shifted to a visibly higher baseline for the entire two-week post-event period.
- **General/Future Volatility:** Broader subreddits (Green lines) exhibit extreme volatility in average scores and comments. This indicates that while general communities react

emotionally (high scores/upvotes) to news, they do not necessarily convert that excitement into sustained discussion volume in the same way focused communities do.

## 3.3 Temporal Analysis

### 3.3.1 Business Question

When is the optimal time to post content on AI-related subreddit to maximize engagement and content quality?

### 3.3.2 Analysis Approach

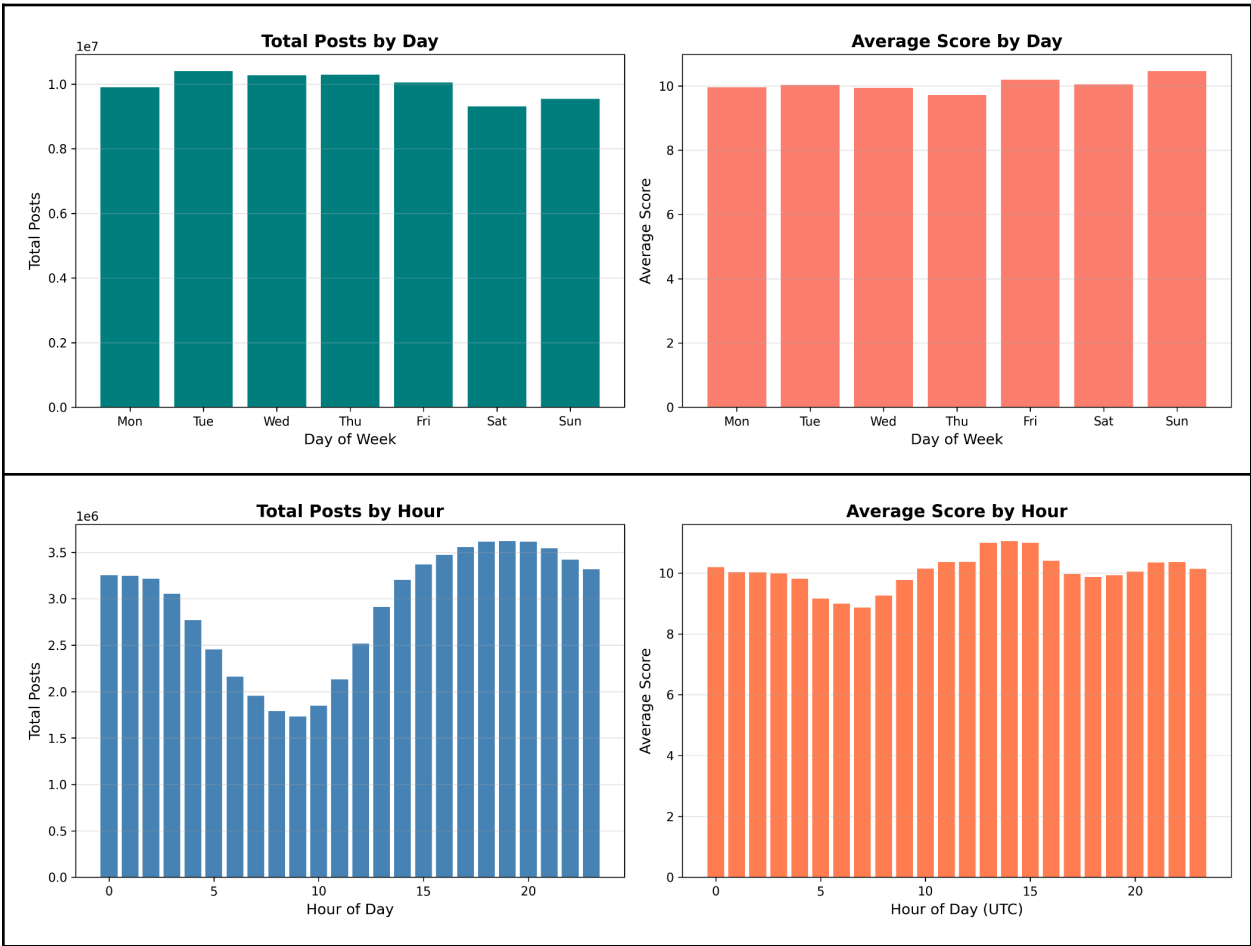
Temporal patterns in engagement were examined by first extracting the `day_of_week` and `hour` variables from each post's `created_utc` timestamp. Using these temporal features, the dataset was aggregated at the `subreddit × day_of_week × hour` level to compute both the total number of posts (`num_posts`) and the average score (`avg(score)`) associated with each time window. These aggregated metrics were visualized through heatmaps and bar charts to reveal variations in posting volume and engagement across different periods of the week. A comparison of the time windows with the highest posting activity and those with the highest average scores enabled the identification of optimal posting periods that balance visibility with engagement quality. This approach provides a systematic framework for determining when posts are most likely to receive attention and achieve higher scores.

### 3.3.3 Posting Patterns by Day and Hour

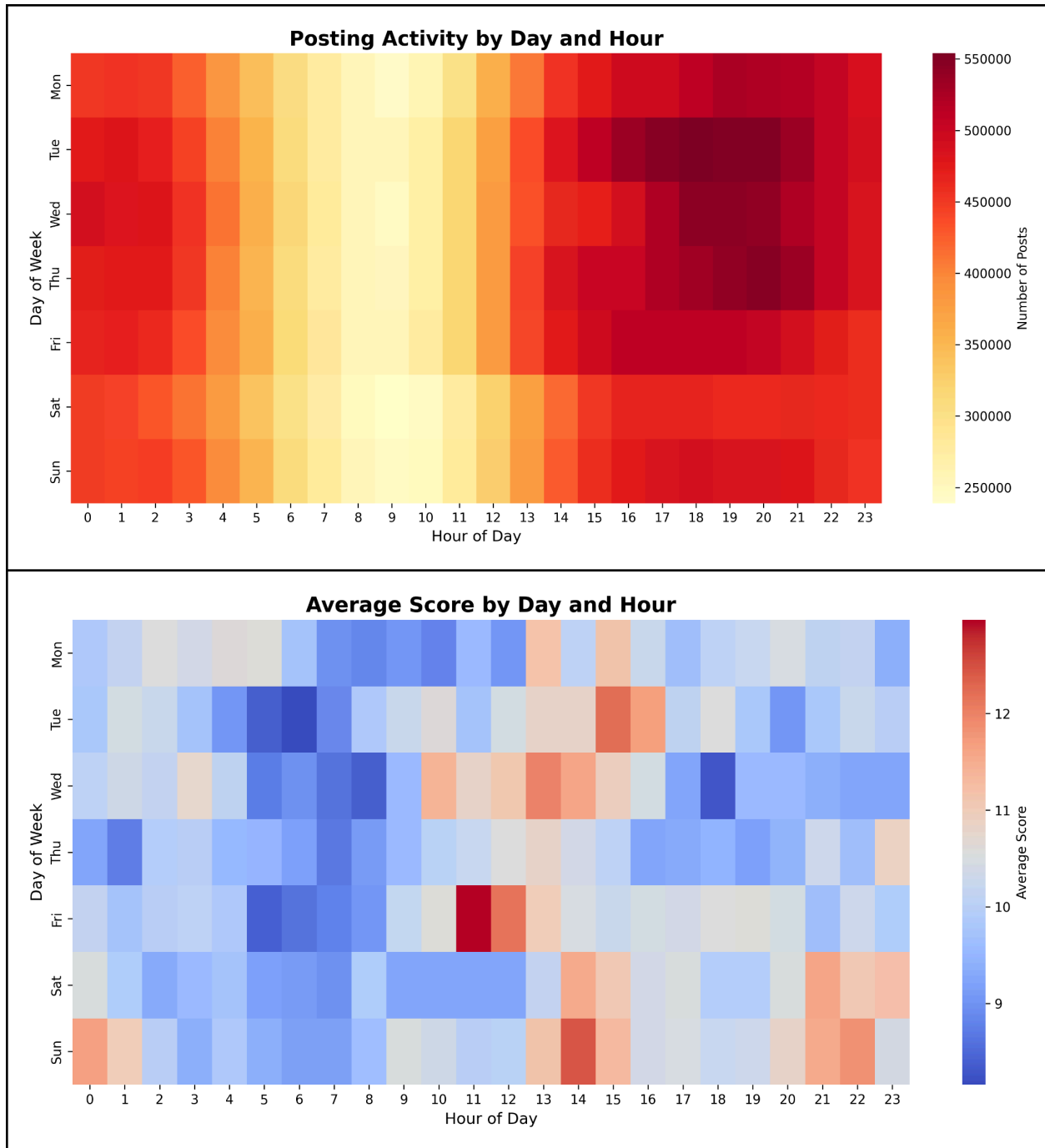
Analysis of posting activity across days of the week reveals a relatively stable distribution of total posts, with slight increases observed from Tuesday to Thursday and a noticeable decline during the weekend. Despite this variation in posting volume, the average score by day exhibits only minor fluctuations, suggesting that the likelihood of receiving higher engagement is not strongly dependent on the day a post is made. Interestingly, Sundays show the highest average score despite having lower overall posting volume, indicating that reduced competition for visibility may enhance engagement quality. This pattern suggests that while weekday activity is generally more frequent, engagement outcomes do not follow the same trend and may favor periods with comparatively lower posting density.

Temporal variation by hour of day shows a more pronounced diurnal cycle. Total posting activity peaks between 15:00 and 21:00 UTC and is lowest between 06:00 and 10:00 UTC, reflecting global patterns of waking hours and online engagement across major English-speaking regions. In contrast, average scores follow a smoother trajectory, with a gradual increase from late morning to early afternoon and a peak between 14:00 and 16:00 UTC. This divergence between posting volume and average score indicates that posts made during periods of moderate activity—rather than during peak posting hours—tend to yield the highest engagement.

Specifically, the mid-afternoon UTC window coincides with both high user activity and relatively less posting saturation, creating an optimal balance between visibility and competition.



Overall, the comparison of these temporal patterns suggests that the most active periods are not necessarily the most rewarding in terms of engagement quality. While posting volume tends to follow predictable weekly and daily cycles, average scores exhibit a more complex relationship with temporal dynamics. Optimal posting windows appear to occur during times when user presence is high but posting competition is not at its maximum—particularly Sunday for weekly patterns and early-to-mid afternoon UTC for daily patterns. These findings highlight the importance of distinguishing between sheer activity levels and actual engagement outcomes when identifying effective posting strategies.



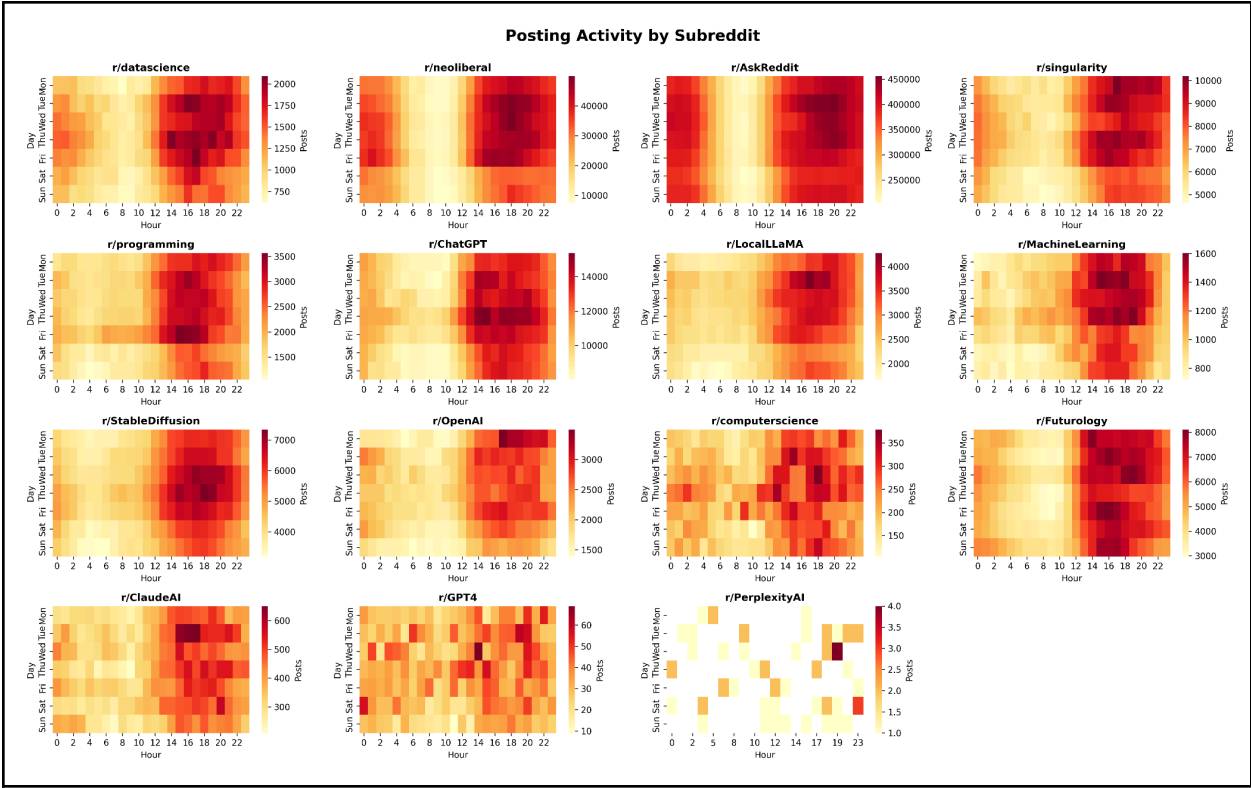
### 3.3.4 Posting Activity by Subreddit

Posting activity exhibits consistent temporal patterns across most AI-related subreddits, with clear peaks during weekday afternoons and evenings (approximately 15:00–22:00 UTC). High-traffic communities such as [r/AskReddit](#), [r/ChatGPT](#), [r/programming](#), and [r/Futurology](#) display particularly pronounced activity surges, reflecting their large user bases and broad topical appeal. Technical subreddits including [r/datascience](#), [r/MachineLearning](#), [r/LocalLLaMA](#),



and r/computerscience show similar diurnal structures but at lower absolute volume, with activity gradually intensifying from morning to late afternoon before tapering off at night. Creative or model-specific communities such as r/StableDiffusion and r/OpenAI also demonstrate strong afternoon-evening peaks, suggesting that both technical work and creative experimentation tend to cluster during shared global working or leisure hours.

Several subreddits display distinctive patterns. r/neoliberal exhibits a uniformly high posting density throughout weekday afternoons, reflecting the integration of AI discussions into broader ongoing political and economic debates. In contrast, r/singularity shows a stable but moderate activity curve, aligning with a smaller but persistent community focused on long-term AI futures. The smallest community, r/PerplexityAI, demonstrates sparse and irregular activity, indicative of its niche user base and emerging adoption stage. Despite these individual differences, the heatmaps collectively reveal a consistent structural trend: across nearly all subreddits, engagement accelerates in the afternoon, peaks between 17:00–21:00 UTC, and declines during early morning hours. These patterns suggest a strong alignment with global user availability and highlight shared behavioral rhythms in online participation across both general-interest and specialized AI communities.



## 3.4 Top Users Across AI Subreddits

Reddit's AI ecosystem is shaped disproportionately by a small group of hyper-active users. These top contributors generate a large share of posts, seed early conversations, and effectively set the tone for entire communities. Understanding who these users are and how they move across subreddits is crucial for mapping how information, norms, and innovations circulate within the AI sphere.

This section focuses on the top 1% of contributors across major AI/tech subreddits. By examining their behaviors — what communities they engage with, how long they stay active, and where their highest-impact content appears — we gain insight into the structural forces driving subreddit growth, cross-community influence, and discourse formation.

### 3.4.1 Business Question

What are the behavioral patterns of the top 1% contributors across the AI/tech sphere?

Specifically, we ask:

#### 1. Are top contributors specialists or generalists?

Do they focus their activity in a single subreddit, or do they spread across multiple AI communities? We categorize users as:

- **Specialists:** 1 subreddit
- **Broad participants:** 2–3 subreddits
- **Generalists:** 4+ subreddits

#### 2. Which subreddit combinations are most frequently co-visited?

These overlap patterns reveal natural clusters and migration pathways — e.g., ChatGPT → LocalLLaMA or MidJourney → StableDiffusion.

#### 3. How long have top contributors been active in the AI ecosystem?

Does high-engagement behavior reflect long-term investment or short bursts during major AI hype cycles?

#### 4. What do top posts look like?

Examining standout posts helps us build intuition about what drives engagement across different AI communities before applying NLP methods.

## 3.4.2 Analysis Approach

### 1. Identifying High-Engagement Users (Top 1%)

Using the combined AI/tech dataset, we calculated each author's total post count across all relevant subreddits (excluding askreddit). Authors at or above the 99th percentile were classified as Top 1% contributors — users who materially shape discourse and community dynamics.

### 2. Measuring Subreddit Participation

For each top contributor, we computed three core participation metrics:

- **Unique subreddits posted in** to map participation breadth
- **Number of subreddits active in** to quantify specialization
- **Most active subreddit** based on post volume or cumulative score
- **Active timespan** from first to last timestamped post

These features enabled us to build contributor profiles that reflect their activity patterns.

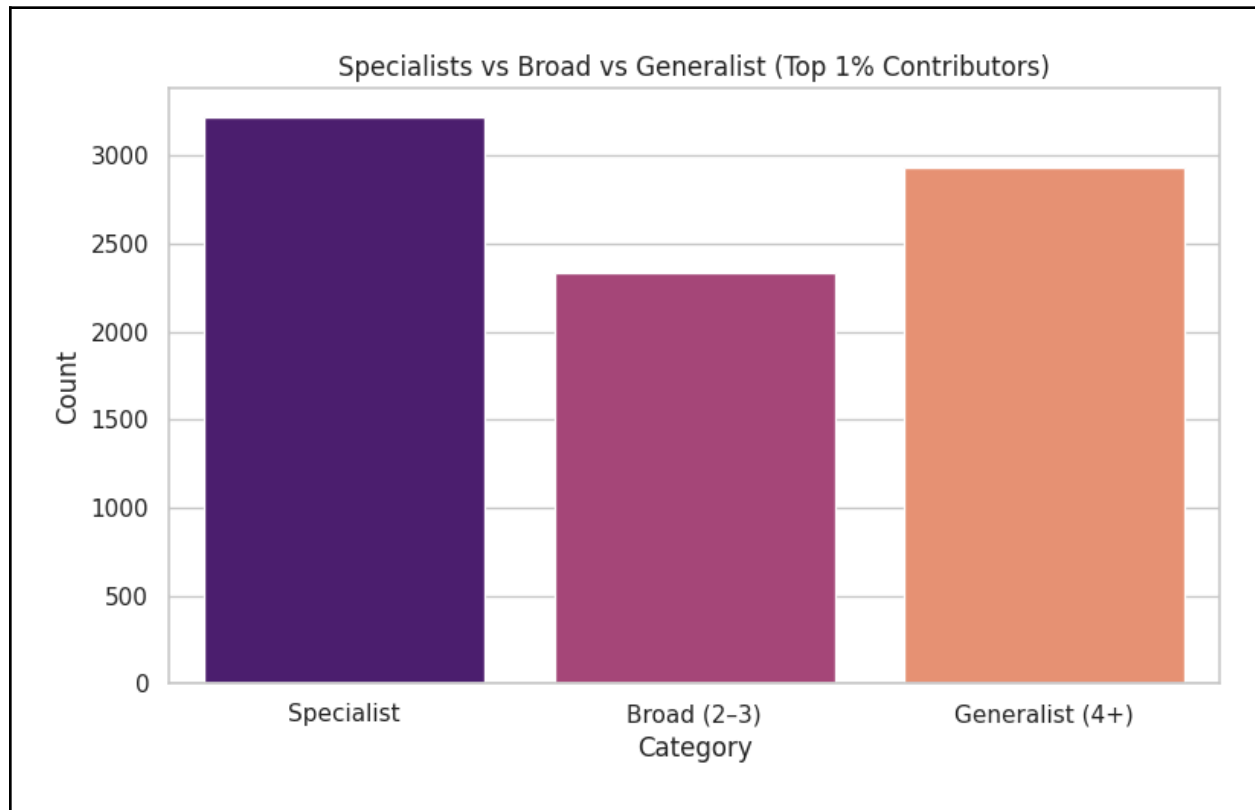
### 3. Classifying Contributors: Specialists vs. Generalists

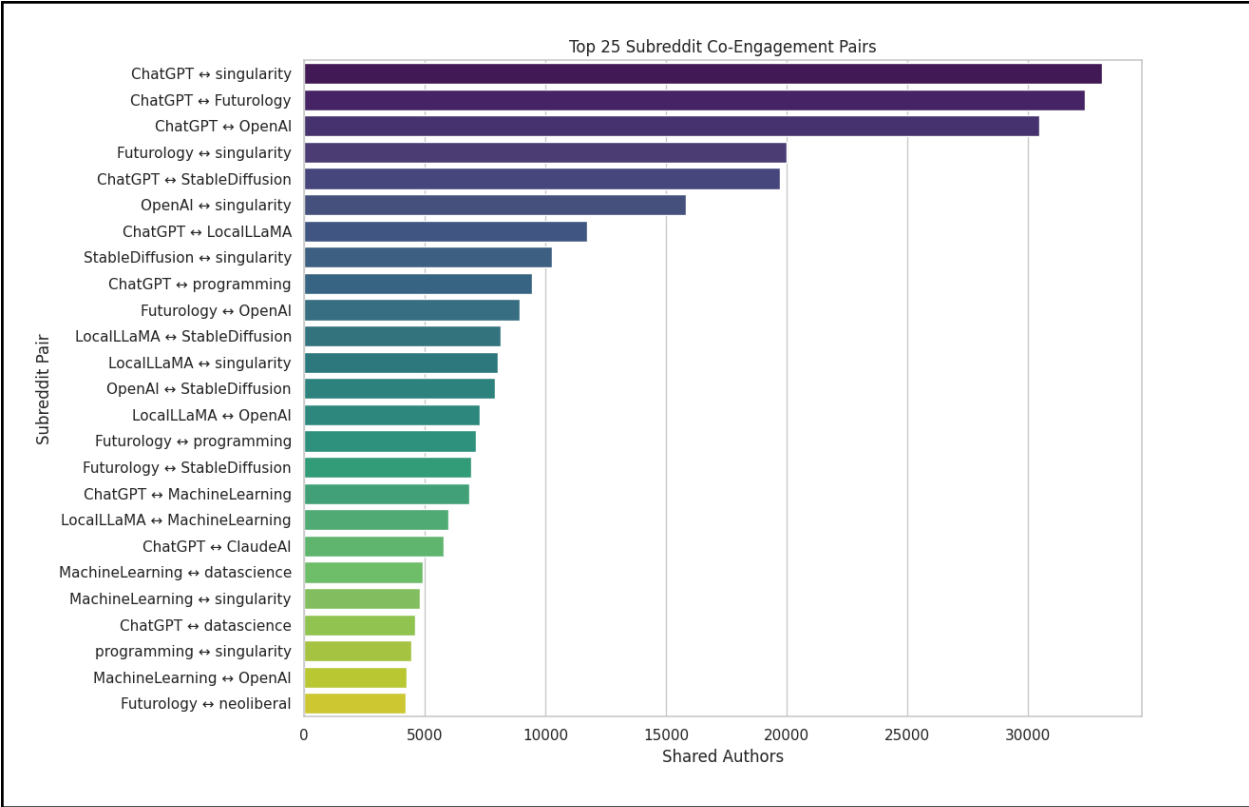
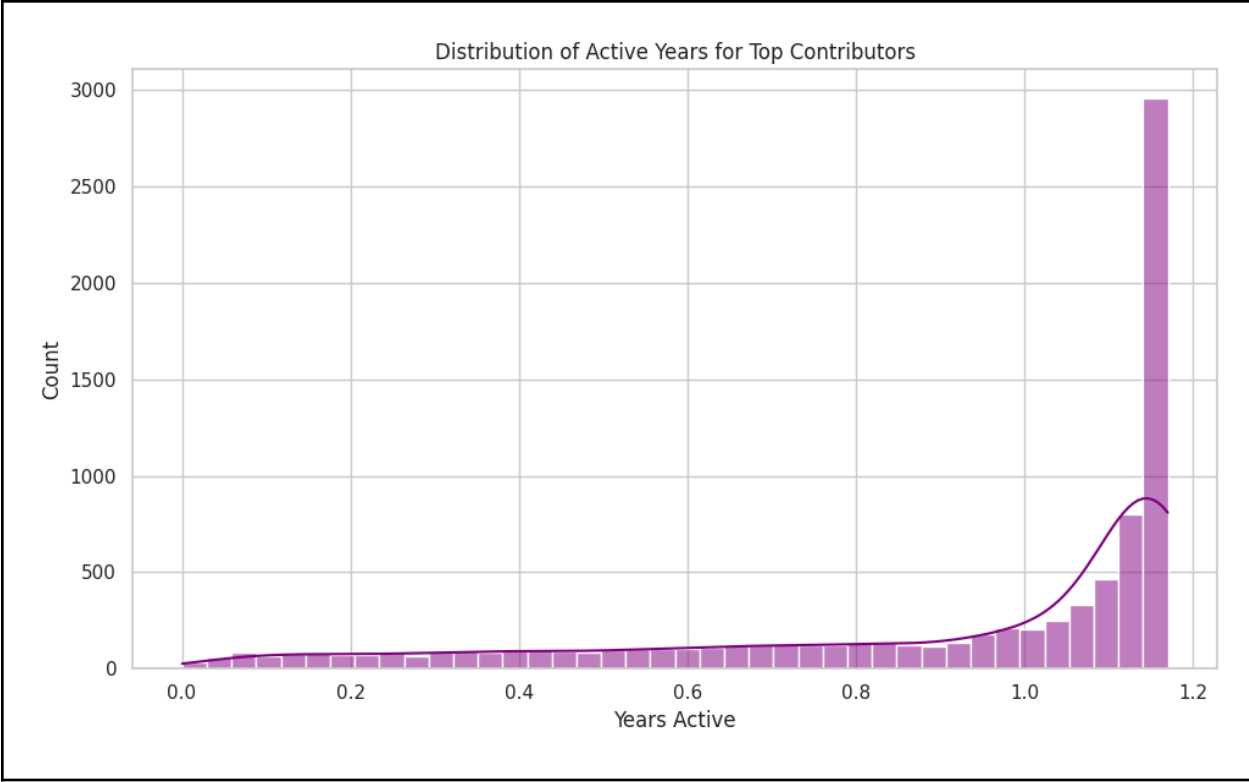
Based on the number of subreddits each contributor participated in, we classified users into behavioral categories:

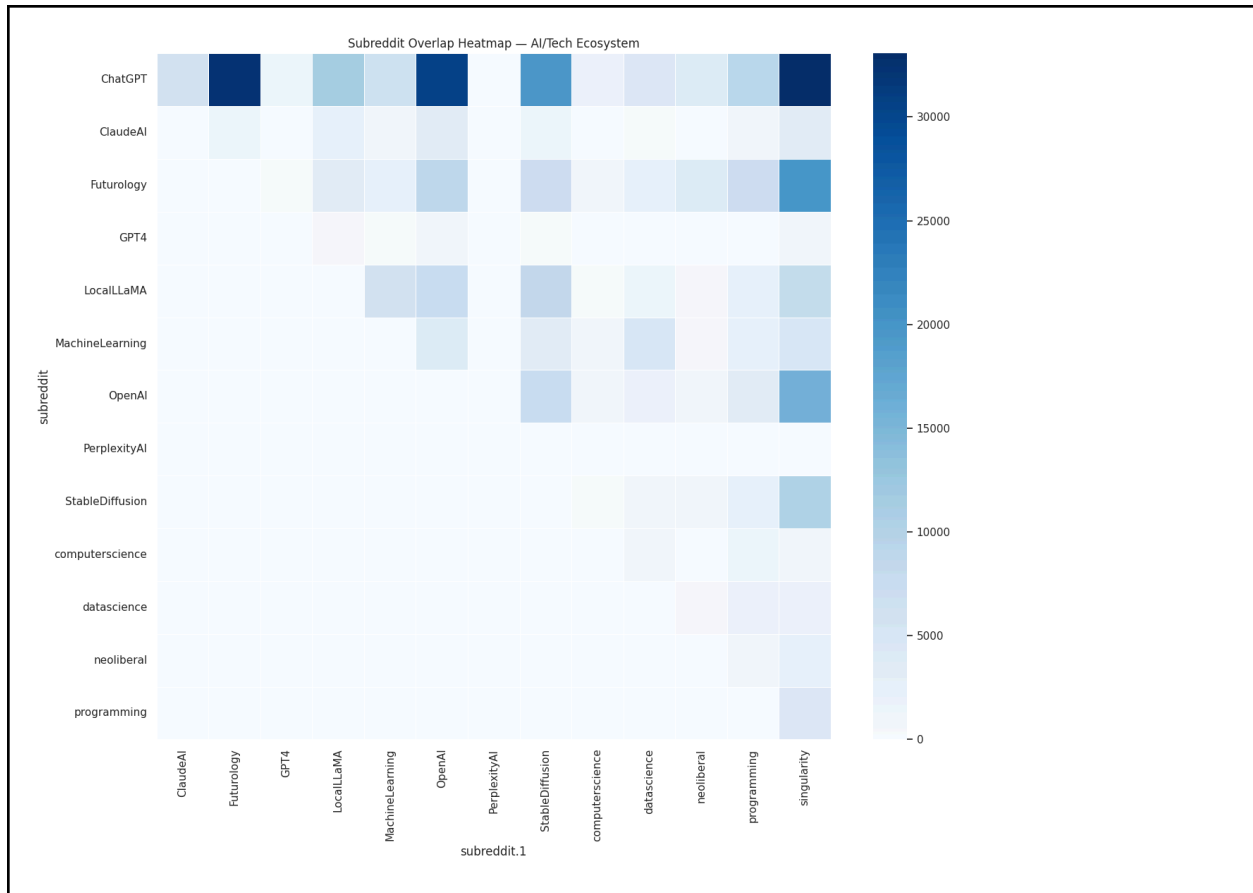
- Specialists: active in exactly 1 subreddit
- Broad participants: active in 2–3 subreddits
- Generalists: active in 4 or more subreddits

This categorization captures whether users cluster tightly around niche communities or move fluidly across multiple AI/tech spaces.

### 3.4.3 Insights







### 3.4.4 Key Observations:

#### 1. Specialists dominate — but generalists play a critical connective role.

Specialists are the largest group, indicating that much of high-volume posting happens within tightly focused communities. Yet generalists form the second-largest group, suggesting a parallel population of “ecosystem roamers” who distribute attention across many subreddits. These generalists often serve as bridges that transport ideas, norms, and debates between communities.

The middle ground — contributors active in 2–3 subreddits — is surprisingly small, reinforcing a polarized behavior: either you specialize deeply or you roam widely.

#### 2. Specialists dominate — but generalists play a critical connective role.

The active-span distribution spikes sharply near the one-year mark. This pattern aligns with the post-2022 acceleration of AI interest: many top contributors are relatively new but extremely active users who surged into the ecosystem during the generative-AI boom.

This suggests that high-engagement behaviors in AI subreddits are tied to rapid cycles of technological novelty rather than long-term community tenure.

### **3. Overlap patterns reveal two major behavioral clusters.**

Subreddit co-engagement and the overlap heatmap show clear community structures:

- A consumer + future-oriented cluster (ChatGPT, OpenAI, GPT-4, Futurology, singularity)
- A technical + builder cluster (LocalLLaMA, MachineLearning, datascience, programming, StableDiffusion)

Strong internal overlaps form within each cluster, while cross-cluster links are weaker. A few “bridge pairs,” like ChatGPT ↔ LocalLLaMA or StableDiffusion ↔ LocalLLaMA, represent user pathways from novice exploration to deeper technical involvement.

### **4. High-engagement posts rely on narrative, novelty, and utility.**

Upon a manual exploration of top posts, across communities, the top posts consistently feature:

- personal discovery stories
- future speculation
- concise tutorials or practical hacks
- culturally resonant humor or commentary

This suggests that what rises to the top is less about technical depth and more about storytelling, clarity, and emotional or conceptual novelty. These patterns provide early intuition for downstream NLP work.

## **4. Natural Language Processing (NLP)**

### **4.1 Exploratory Topic Modeling**

#### **4.1.1 Business Question**

What are the dominant topics discussed in AI-related Reddit posts?

#### **4.1.2 Analysis Approach**

To examine thematic patterns in AI-related discourse at scale, the dataset was first filtered to include comments from major AI-focused subreddits, while removing empty posts and those with fewer than 30 characters to ensure substantive content. Text underwent a multi-stage preprocessing pipeline, including tokenization, lowercasing, expanded stopword removal, and a custom UDF designed to filter out URLs, Reddit-specific artifacts, non-alphabetic strings, and

very short tokens. Documents with no remaining tokens were excluded. Cleaned tokens were then converted into a sparse term–document matrix using a CountVectorizer with a vocabulary cap of 10,000 and a minimum document frequency of 50. LDA topic models with 15 topics were trained to capture latent themes within the conversations. Because the neoliberal subreddit contains a substantial amount of politically oriented discourse that could disproportionately influence the learned topic structure, two separate models were estimated: one excluding this subreddit to isolate AI-focused discussions, and a second including it to examine how politically charged content reshapes or broadens the thematic landscape. This dual-model design enabled a clearer decomposition of core AI topics and the additional ideological framing introduced by political communities.

#### 4.1.3 Discovered Topics without neoliberal subreddit

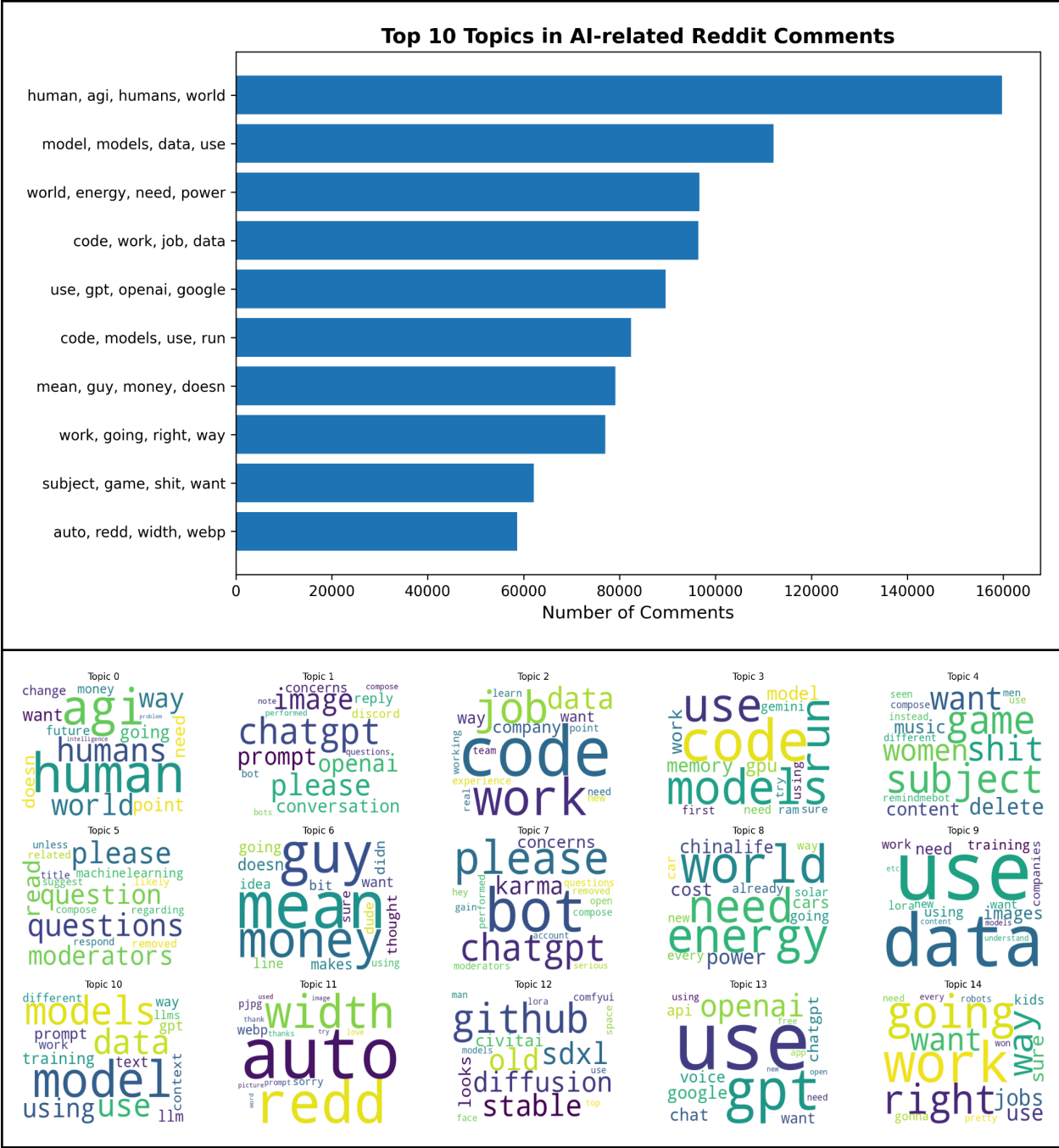
##### Topic 0 — AGI, Humanity, and Long-Term Societal Impact

This topic centers on abstract and philosophical discussions about artificial general intelligence (AGI) and its potential consequences for humanity. Users debate existential risks, the future role of humans in an AI-driven world, ethical dilemmas, and broad societal transitions. The language shows a mixture of concern, speculation, and curiosity about how intelligence and power may evolve as AI systems advance.

##### Topic 1 — ChatGPT Usage, Prompting, and Interaction Issues

This topic reflects users' practical experiences interacting with ChatGPT and similar conversational systems. Discussions revolve around prompt engineering, image generation requests, response quality, clarifications, and troubleshooting. Many posts involve seeking help to improve outputs or resolve unexpected behaviors in conversations with ChatGPT.





Topic 2 — Jobs, Workflows, and Practical Coding Questions

This topic captures career-oriented and practical conversations related to programming, employment, workplace tasks, and gaining experience in tech fields. Users share advice on job responsibilities, onboarding, real-world coding expectations, team dynamics, and skill development. It reflects a blend of professional support and technical guidance.

Topic 3 — Model Execution, Hardware Constraints, and Troubleshooting

This topic includes detailed technical discussions about running AI models locally or in the cloud. Posts frequently mention GPU limits, RAM shortages, memory errors, and runtime issues when working with large models such as Gemini or other LLMs. It represents highly technical troubleshooting and optimization conversations.

#### Topic 4 — Entertainment Content, Gaming, and Moderation Issues

This topic combines discussions about gaming, entertainment content, and moderation-related matters on Reddit. Users comment on games, music, gender-related disputes, and general content management. The presence of strong language and topic diversity suggests a more socially-oriented and sometimes contentious theme.

#### Topic 5 — Q&A, Moderation, and MachineLearning Subreddit Meta-Discussions

This topic represents posts directed toward subreddit moderators or users seeking clarification on posting rules. It includes questions about removed content, how to properly format inquiries, and issues specific to the MachineLearning subreddit. The discussion style is formal and administrative, focusing on community governance.

#### Topic 6 — Casual Commentary, Opinions, and Informal Dialogue

This topic consists of informal and conversational exchanges involving personal opinions, light humor, reactions, and small disagreements. The vocabulary (“guy,” “dude,” “mean,” “thought,” “sure”) reflects a casual tone rather than technical or analytical subject matter. It indicates everyday community engagement rather than AI-focused content.

#### Topic 7 — Bot Activity, Karma, and Moderation Interactions

This topic focuses on interactions involving Reddit bots, karma mechanics, account issues, and moderation concerns. Users often request help when posts are removed or bots behave unexpectedly. It reflects platform-level discussions rather than AI-specific technical content.

#### Topic 8 — Energy Consumption, Infrastructure, and Global AI Costs

This topic highlights large-scale concerns about the energy requirements of AI systems. Discussions include power consumption, hardware costs, environmental impact, solar energy, and geopolitical dynamics (e.g., China’s role in technological development). The posts suggest public interest in the sustainability and resource demands of AI at scale.

#### Topic 9 — Data, Images, Training Pipelines, and Model Fine-Tuning

This topic captures technical conversation about data pipelines, image datasets, model training, and fine-tuning techniques such as LoRA. Users discuss how models use training data, how images are processed, and how AI systems can be adapted to new tasks. It represents hands-on ML engineering practice.

#### Topic 10 — LLM Fundamentals: Models, Training, Text, and Context Handling

This topic involves theoretical and conceptual conversations about large language models (LLMs). Users discuss how models handle context, how they are trained, the role of text data, and differences between model architectures. It reflects more foundational AI knowledge compared to the practical focus of Topic 9.

#### Topic 11 — Image File Formats, Compression, and Output Issues

This topic consists of posts addressing image formats (webp, png), file width/size, and how to convert or manage images produced by generative models. Many users seek help resolving compatibility issues or adjusting model outputs. It is highly technical but focused on imaging workflows rather than LLMs.

#### Topic 12 — Stable Diffusion Ecosystem, Tools, and Model Assets

This topic is strongly connected to the Stable Diffusion community. Users discuss SDXL models, ComfyUI workflows, LoRA assets, and tools hosted on GitHub or Civitai. The content reflects the creative and technical aspects of image-generation ecosystems, including model customization and asset management.

#### Topic 13 — GPT/LLM Applications, APIs, and Integration into Tools

This topic centers on building applications with GPT models, using OpenAI/Google APIs, integrating voice or chat features, and developing new AI-driven tools. Users frequently ask for implementation advice, best practices, and examples. It illustrates the growing interest in applied AI development.

#### Topic 14 — Automation, Robotics, and the Future of Work

This topic explores the societal and labor impacts of automation and robotics. Users debate how AI will reshape jobs, affect future generations, and change workplace expectations. The tone mixes economic concern with general curiosity about technological transformation.

### 4.1.4 Discovered Topics with neoliberal subreddit



### Topic 1 — ChatGPT Prompting, Image Requests, and User Support

This topic features hands-on questions and troubleshooting related to ChatGPT. Users ask how to craft effective prompts, generate images, improve responses, and resolve issues encountered during conversations with the model. Many posts involve seeking help, sharing concerns, and asking for clarifications regarding ChatGPT or OpenAI services.

### Topic 2 — Historical, Legal, and State-Level Discussions Mixed With Political References

This topic blends broader societal discussions—including history, law, factual interpretation, and state-level issues—with occasional references to U.S. political figures such as Biden. The vocabulary signals debate-style interactions where users argue about facts, legal interpretations, and the “state of things.” The tone is analytical and sometimes contentious.

### Topic 3 — Language, Meaning, and Conceptual Interpretation

This topic centers on meta-level discussions about language, writing, meaning, and conceptual points. Users debate how words should be interpreted, how language shapes understanding, and how ideas are communicated. Many posts are reflective or philosophical, discussing writing habits, learning processes, and differences in communication styles.

### Topic 4 — Serious Topic Discussions: Climate, Energy, Nuclear Issues, and Social Change

This topic brings forward serious discussions about societal challenges such as climate change, nuclear power, energy transitions, and social issues involving children or attention dynamics. Users express gratitude, debate policy or ethics, and exchange views on significant global or societal concerns. The tone is more earnest compared to other topics.

### Topic 5 — Economic and Corporate Power: Money, Companies, and AGI Impacts

This topic highlights discussions about the economic implications of AI, the power of corporations, and the relationship between AGI, money, and society. Users debate whether companies will dominate AI development, how economic incentives shape innovation, and what the future might look like as AGI progresses. It reflects the neoliberal subreddit’s influence, introducing economic framing and concerns about corporate decision-making.

### Topic 6 — U.S. Electoral Politics, Parties, and Polarization

This topic is explicitly political, dominated by mentions of Trump, Biden, elections, voters, and party dynamics. Users argue about government decisions, presidential candidates, and ideological divisions. This political framing appears only in the neoliberal-included model and highlights the strong overlap between AI discussions and partisan debates within those subreddits.

### Topic 7 — Human Nature, Consciousness, and Cognitive Questions

This topic includes philosophical and cognitive discussions about the human mind. Users debate consciousness, the brain, identity, belief, and what makes humans unique. Many posts attempt to define or compare human mental processes to artificial intelligence, reflecting public fascination with cognition and selfhood.

#### Topic 8 — Work, Data Skills, Learning, and Human Intelligence

This topic reflects conversations about data-related jobs, learning, training, and work experience. Users ask for advice on job preparation, data science learning paths, and skill-building. It blends human intelligence, training difficulty, and career development in technical fields.

#### Topic 9 — Global Politics, China, the World Economy, and Demographic Concerns

This topic highlights geopolitical discussions, especially about China, world events, global population issues, the housing market, and economic pressures. Users discuss national-level AI competition, demographic changes, and socioeconomic inequalities. It becomes more global and macroeconomic compared to the “without neoliberal” model.

#### Topic 10 — Using Models for Art, Images, and Creative Purposes

This topic focuses on applying models for artistic and image-generation purposes. Users ask how to use models for art, generate images, adjust prompts, and experiment creatively. The tone is practical but creative, representing the AI art community’s workflows.

#### Topic 11 — Image Formats, File Issues, and International Conflicts

This topic is unique because it merges two unrelated clusters that co-occur in the dataset: (1) technical issues with image formats (webp, pjpeg, width), and (2) political conversations referencing Israel, Gaza, Hamas, Ukraine, and military conflict. This suggests that neoliberal-included spaces frequently discuss international conflicts, causing the LDA model to merge global politics with unrelated image-format terms due to overlapping vocabulary.

#### Topic 12 — Gender, Media Dynamics, and Social Commentary

This topic involves heated discussions about men, women, gender roles, media narratives, and social behavior. The presence of emotionally charged language reflects disagreements or debates about gendered social interactions. This topic emerges more strongly when neoliberal content is included, indicating a shift toward broader social commentary.

#### Topic 13 — Moderation Issues, Neoliberal Subreddit Processes, and Community Governance

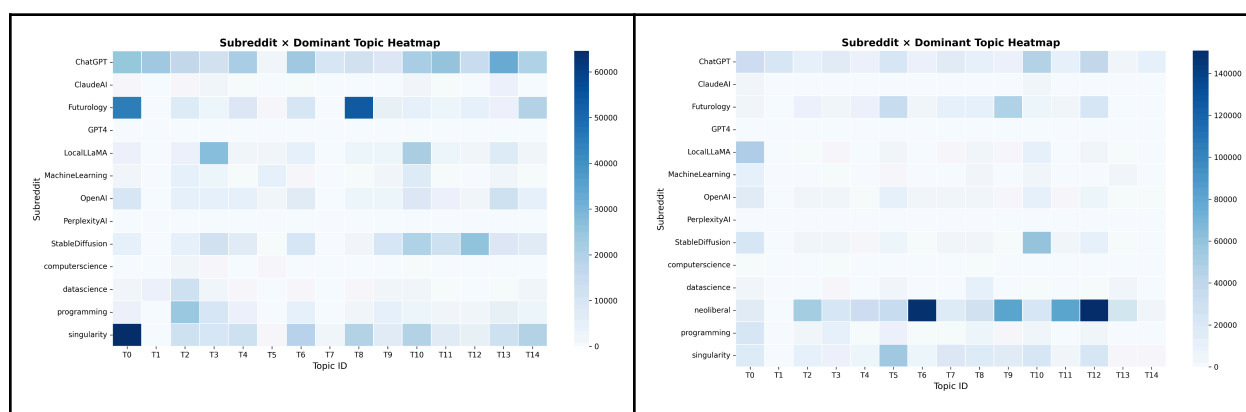
This topic directly reflects the influence of the neoliberal subreddit. Users reference moderation, post submissions, participation rules, removed content, and community governance. The explicit keyword “neoliberal” demonstrates that this space shapes user discussions about policy, moderation fairness, and managing discourse quality.

## Topic 14 — Bots, Karma, Voice Features, and Platform-Level Concerns

This final topic concerns Reddit mechanics—bots, karma, account rules—and ChatGPT’s voice or interactive capabilities. Users ask about permissions, technical limitations, and posting etiquette. It reflects platform-level engagement more than AI-specific content.

### 4.1.5 Subreddit–Topic Distribution Comparison (Without vs. With neoliberal)

Including the neoliberal subreddit produces clear shifts in the dominant topic distribution across communities. In the model without neoliberal, topic prevalence is primarily driven by technical or model-specific subreddits (e.g., ChatGPT, Futurology, LocalLLaMA, StableDiffusion), revealing a structure centered on AI usage, coding, and model functionality. However, in the model with neoliberal, Topic 6 and Topic 12 become heavily amplified due to the strong political and economic discourse characteristic of the neoliberal community. This redistribution highlights how the introduction of a politically oriented subreddit can reshape the overall thematic landscape, increasing the prominence of socio-political themes and reducing the relative influence of purely technical communities in the aggregated topic structure.



## 4.2 Emotion & Sentiment Patterns of High-Engagement Users

### 4.2.1 Business Question

How do high-engagement contributors (top 1% posters) emotionally express themselves across AI and tech subreddits, and what differences emerge between technical, creative, and general-interest AI communities?

### 4.2.2 Analysis Approach

This analysis explores the emotional tone of Reddit’s most active AI and technology contributors by focusing on the top one percent of users across the platform. Rather than examining only what these power users discuss, the goal is to understand how they express themselves emotionally and how their tone differs across AI-oriented communities.

To identify these voices, we first computed the global 99th-percentile threshold for posting activity and collected all comments and submissions made by users above that cutoff. After gathering their content, we applied a comprehensive preprocessing pipeline that included lowercasing, removing URLs, stripping markdown elements, tokenizing text, filtering stopwords, and cleaning Reddit-specific artifacts such as flairs, user tags, and formatting noise.

Each processed text was then classified using a multi-category emotion lexicon covering joy, love, anger, fear, sadness, surprise, and neutrality. We calculated proportional emotion scores for each subreddit and normalized the distributions to account for differences in community size, allowing for meaningful comparisons across diverse spaces.

The emotional patterns were then compared across several thematic clusters:

- Text AI communities such as ChatGPT, OpenAI, GPT-4, ClaudeAI, and PerplexityAI
- Creative AI communities including StableDiffusion, MidJourney, Sora, and AIArt
- Research and technical communities like MachineLearning, datascience, programming, and computerscience
- Futures-oriented and ideological communities such as Futurology, singularity, and neoliberal with AskReddit serving as a baseline for general online sentiment.

Finally, radar-chart visualizations were used to highlight the distinctive emotional “profiles” of each community. These visual summaries reveal which groups express relatively higher levels of fear, anger, joy, neutrality, or surprise, offering insight into how different corners of Reddit emotionally navigate the rapid evolution of AI.

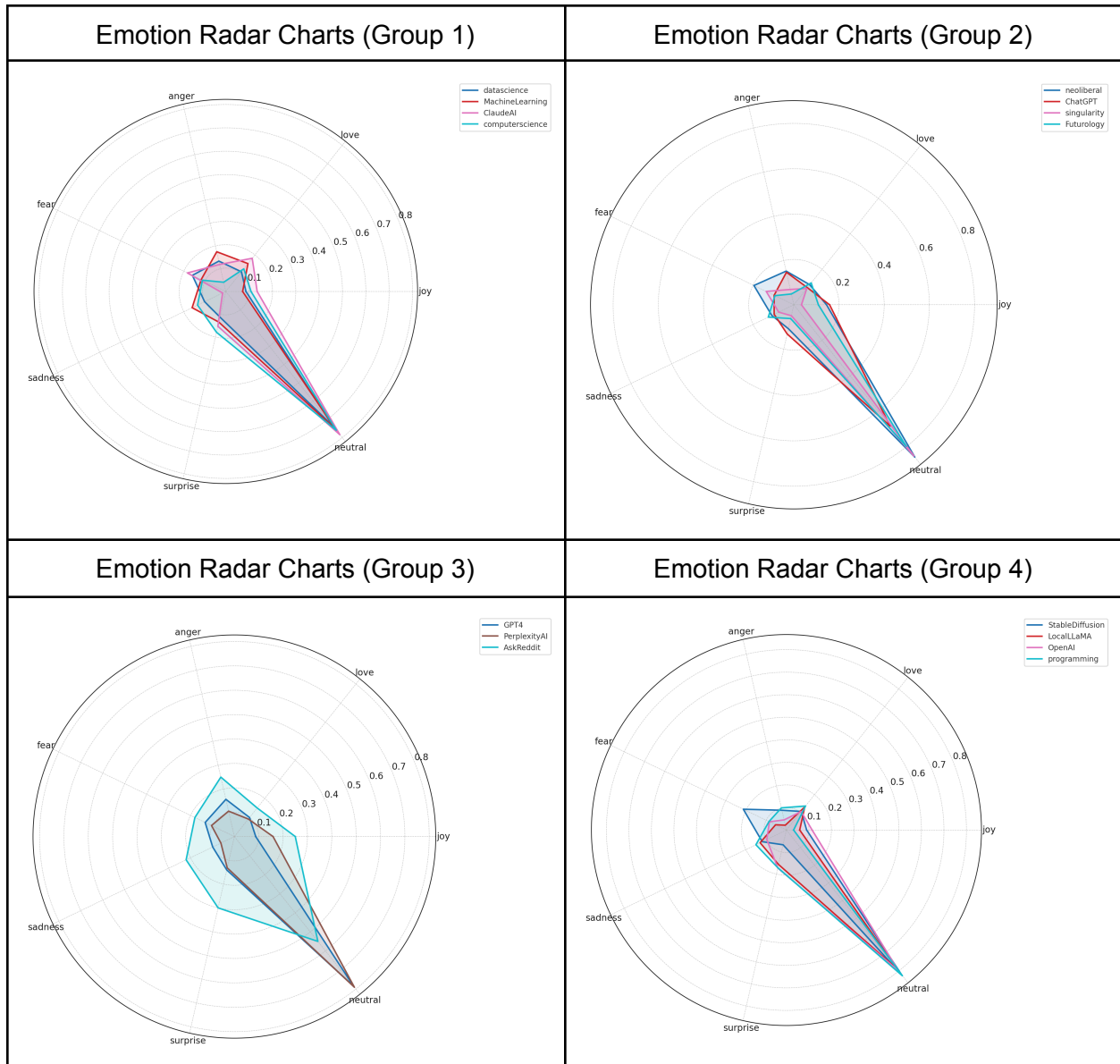
### 4.2.3 Overall Sentiment Patterns Among High-Engagement Users

Across all AI-related subreddits, neutral sentiment overwhelmingly dominates. The radar charts consistently show that top contributors communicate in a highly analytical, information-driven style with minimal emotional expression. This suggests that power users approach AI primarily as a technical or problem-solving domain rather than an emotionally charged topic.

Technical subreddits, such as MachineLearning, datascience, programming, computerscience, and LocalLLaMA, show the flattest emotional profiles of all groups. Joy, fear, anger, and sadness remain extremely low, reflecting discussions focused on debugging, model evaluation, optimization, and tool deployment rather than personal reactions or subjective interpretations.

This overall neutrality indicates a shared communication norm among expert-level or highly active users: precision over emotion, information over affect.





#### 4.2.4 Community-Specific Emotional Variation

While technical communities show minimal emotion, creative AI communities (StableDiffusion, MidJourney, AI Art) exhibit greater emotional variability, especially in surprise, anger, and fear. These emotional spikes match the nature of discussion: debates around creativity, copyright, ethics, and model instability often trigger more expressive reactions.

In contrast, future-oriented and ideological communities—including Futurology, singularity, and neoliberal—display higher levels of fear, sadness, and anger. These patterns reveal deeper societal anxieties about automation, labor displacement, existential risk, and political-economic

implications of AI. This group interprets AI not as a technical tool but as a transformative and potentially destabilizing force.

Together, these patterns show that the emotional tone of AI discourse varies systematically by community purpose:

- Technical → neutral, constrained
- Creative → expressive, debated
- Futures/ideology → concerned, anxious

#### 4.2.4 Model-Specific vs General Public Emotional Profiles

Model-focused subreddits—ChatGPT, ClaudeAI, GPT4, OpenAI, PerplexityAI—form a tight emotional cluster. Their sentiment profiles are almost identical, emphasizing utility-driven interactions. Users here engage with tools pragmatically, discussing prompts, output quality, and capabilities without strong emotional tone.

On the opposite end, AskReddit, used as the general-public baseline, shows the highest emotional expression overall. Conversations about AI among everyday users contain significantly more fear, sadness, and uncertainty. This contrast highlights a key divide between expert and non-expert communities:

- Power users: stable, neutral, tool-focused
- General public: uncertain, emotionally reactive to AI's rapid evolution

This divergence suggests that emotional reactions to AI are not uniformly distributed across the internet; they depend heavily on user expertise, community norms, and the framing of AI within each subreddit.

### 4.3 Sentiment Analysis

#### 4.3.1 Business Question

How has sentiment toward different generative AI tools (e.g., ChatGPT, Midjourney, Sora) shifted over time across subreddits?

#### 4.3.2 Analysis Approach

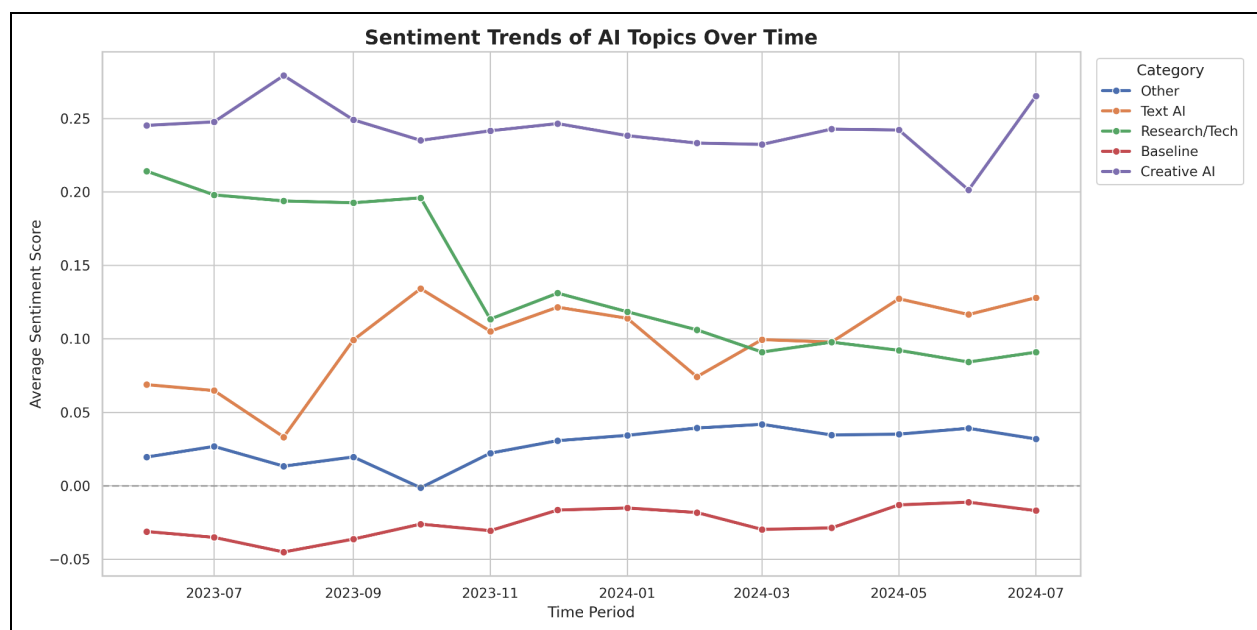
To quantify emotional shifts within community discussions, this study employs a **lexicon-based sentiment analysis framework**. The methodology consists of four distinct phases:

1. **Text Preprocessing & Feature Extraction:**

- Raw text data was processed using the **Spark ML Pipeline**. Key steps included tokenization, stop-word removal, and extensive cleaning to strip URLs, Reddit-specific formatting, and Markdown artifacts.
  - Feature extraction was performed using **HashingTF** (5,000 features) to convert text into numerical vectors for efficient processing.
2. **Lexicon-Based Sentiment Scoring:**
- Sentiment was calculated using a polarity-based lexicon approach, mapping words to positive and negative values.
  - Composite scores were normalized on a scale of **-1.0 (Very Negative) to +1.0 (Very Positive)**, accounting for the ratio of positive-to-negative tokens within each document.
3. **Categorical Stratification:**
- Discussions were classified into four domain-specific clusters to isolate trends:
    - **Text AI:** (e.g., ChatGPT, GPT-4, Claude, Perplexity).
    - **Creative AI:** (e.g., Midjourney, Stable Diffusion, Sora).
    - **Research/Tech:** (e.g., r/MachineLearning, r/ArtificialIntelligence).
    - **Baseline:** (General r/AskReddit threads, serving as a control group).
4. **Temporal Aggregation:**
- Scores were aggregated monthly by category to calculate Mean Sentiment Score (MSS) and Standard Deviation. These metrics were cross-referenced with **Post Volume** to identify correlations between discussion intensity and community sentiment from June 2023 to June 2024.

### 4.3.3 Findings

The longitudinal analysis reveals a clear stratification in how different AI applications are perceived by their respective communities.

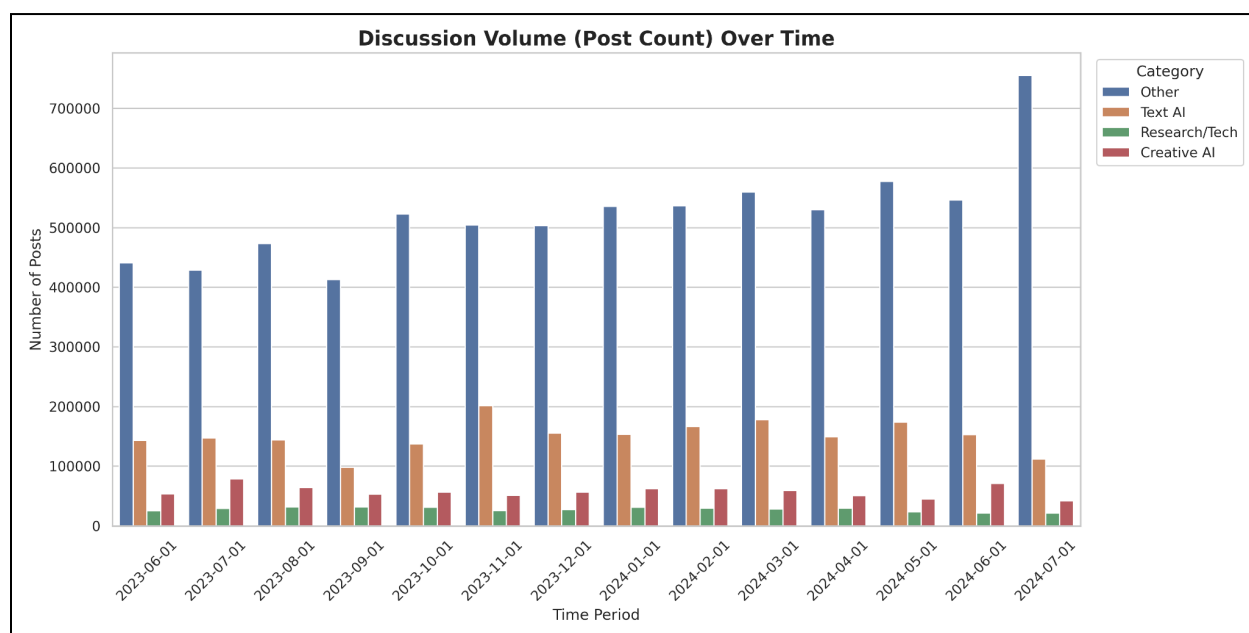


**1. The "Creative Satisfaction" Gap** Creative AI (Purple Line) consistently maintains the highest sentiment scores, fluctuate between **0.20 and 0.28**.

- This suggests that users of generative image/video tools experience higher immediate gratification and lower frustration compared to other groups.
- The visual nature of these communities fosters a "show-and-tell" culture that is inherently positive, keeping sentiment high even when discussion volume fluctuates.

**2. The Research "Reality Check"** Research/Tech sentiment (Green Line) displayed a distinct downward correction.

- In mid-2023, sentiment was high (~0.21), comparable to Creative AI. However, around **October 2023**, there was a sharp decline to **~0.11**, after which it converged with Text AI levels.
- This likely marks the transition from the initial "hype phase" of theoretical excitement to a "utility phase" where practical limitations and technical challenges became the primary focus of discussion.



**3. The Inverse Relationship Between Volume and Sentiment** By comparing the **Sentiment Trends** chart with the **Discussion Volume** chart, a notable inverse pattern emerges.

- **Text AI** (Orange) commands the second-highest discussion volume (averaging ~150k posts/month) but maintains only moderate sentiment (**0.05 – 0.13**).
- High-volume categories tend to be saturated with troubleshooting, bug reports, and usage inquiries, which naturally drags down the average sentiment score compared to niche, lower-volume creative communities.

**4. Specialized Communities vs. The General Public** All specific AI categories (Creative, Text, Research) track consistently higher than the **Baseline** (Red Line), which remains negative (< 0.00).

- This confirms that AI-specific subreddits act as "optimism hubs," whereas general public discourse (represented by Baseline) remains skeptical or critical of the technology.

## 4.4 Profanity Language Frequency Analysis

### 4.4.1 Business Question

What percentage of profanity words is used to describe AI across different subreddits?

### 4.4.2 Analysis Approach

The analysis utilized the [Google Profanity Words dataset](#), a widely adopted, community-maintained list of offensive and inappropriate terms. With 676 stars and 249 forks on GitHub, this dataset demonstrates strong reliability and broad validation across production-grade applications.

The dataset offers several advantages:

- Open-source with transparent inspection and reproducibility
- Actively maintained with frequent updates and community oversight
- Language-specific lists suitable for Reddit text analysis
- Designed for profanity filtering, aligning with toxicity detection goals

### 4.4.3 Processing Pipeline

The analysis followed these steps:

1. Cleaned text by removing URLs, markdown, usernames, and very short posts
2. Applied Spark NLP preprocessing including tokenization, normalization, stop-word removal, and lemmatization
3. Computed profanity metrics per subreddit using Spark-native token filtering
4. Identified top profanity words by counting token frequencies across subreddits
5. Generated visualizations including profanity rate bar charts and per-subreddit word clouds using 30% sampled comments

### 4.4.4 Profanity Patterns Across Subreddits

The results reveal a clear pattern: AskReddit leads with the highest profanity percentage at 8.22%, followed by neoliberal (7.76%) and Futurology (6.56%), signaling more emotionally charged or socially polarized conversations. Meanwhile, subreddits focused on engineering or research show lower profanity levels between 4% and 4.6%.

PerplexityAI notably has the lowest rate at 1.49%, indicating highly utility-driven, information-focused discussions with minimal emotional or toxic language. Overall, the results

[illegible]

Across the word clouds, the most prominent terms include problem, issue, error, wrong, concern, hard, limit, and bad. These words dominate nearly every community, suggesting a shared pattern where users frequently describe difficulties with AI tools, unexpected behaviors, performance failures, and constraints that hinder workflows.

- **GPT4, ChatGPT, ClaudeAI, LocalLLaMA:** Highlight words like wrong, error, bad, break, and slow, pointing to reliability and accuracy problems
- **Futurology and AskReddit:** Show larger emotional expressions like hard, kill, limit, and cheap, reflecting broader worries or sensational narratives about AI
- **PerplexityAI and MachineLearning:** Maintain smaller, more technical clouds focused on concrete issues rather than emotional or vulgar expressions

Overall, the word clouds reveal that core vocabulary across communities centers on problems, errors, and concerns, painting a picture of widespread user frustration and perceived limitations of current AI tools.

## 5. Machine Learning (ML)

### 5.1 Score prediction

To address the first business question, the task was formulated as a regression problem with the target variable defined as the comment score. Model performance was evaluated using RMSE, MAE, and  $R^2$ .

#### 5.1.1 Feature Engineering

The prediction models were trained using a combination of text-based, topic-based, and metadata features.

Text-based features (TF-IDF):

Comment text was lowercased, tokenized using a `RegexTokenizer`, and cleaned with a customized `StopWordsRemover` that removed both standard English stopwords and web-related tokens (e.g., `http`, `jpg`, `youtube`). Tokens were converted into term-frequency vectors via `CountVectorizer` and transformed into TF-IDF features. Because Lasso Regression was used, each word received an interpretable coefficient indicating whether it increases or decreases the predicted score.

Topic-based features (LDA topic distribution):

A pre-trained LDA model provided a `topicDistribution` vector for each comment. These features capture higher-level semantic tendencies (e.g., technical discussion, policy debates, community norms).

Metadata:

From the `created_utc` timestamp, additional contextual features were derived:

- `day_of_week`
- `hour`
- `text_length`

These variables were combined with text and topic features through a `VectorAssembler` to construct the full feature vector.

#### 5.1.2 Model Performance

Three Lasso Regression models (`regParam` = 0.05, 0.10, 0.20) and a Gradient Boosted Trees model were evaluated. Lasso Regression yielded RMSE values around 40–41 with slightly negative  $R^2$  scores, indicating limited predictive power. Gradient Boosted Trees reduced RMSE

to 19.25 and MAE to 6.02, though  $R^2$  remained slightly negative. Overall, the models captured broad trends but struggled to predict exact scores, reflecting the inherently noisy, volatile nature of Reddit engagement.

### 5.1.3 Words Associated With Lower Predicted Scores (Lasso Coefficients)

Across both models—with and without the neoliberal subreddit—Lasso regression assigns the most negative coefficients to conversational, low-information, or meta-discussion terms. Words such as “thanks,” “contact,” “downvoted,” “yeah,” “agree,” “talking,” “subreddit,” “comment,” and “mean” consistently reduce the predicted score of a post.

These patterns suggest that comments containing polite acknowledgments, back-and-forth conversational markers, or administrative/meta-level references are less likely to receive high engagement. Such language typically signals comments that are reactive, routine, or peripheral to the main discussion, which do not attract strong community voting.

When neoliberal is included, additional negatively associated words—such as “compose,” “replies,” “linked,” “depends,” “humans,” and the anomalous numeric token “3024”—suggest the model captures a broader range of conversational or thread-management content that tends to correlate with lower engagement in politically active or highly interactive communities.

### 5.1.4 Words Associated With Higher Predicted Scores (Lasso Coefficients)

The highest positive coefficients exhibit a contrasting pattern. In the without neoliberal model, words such as “survived,” “forums,” “millennials,” “delve,” “youth,” “sleeping,” “detecting,” “armor,” “babies,” “backlash” strongly increase predicted score. Many of these terms are associated with:

- storytelling or narrative framing (survived, youth, millennials)
- emotionally charged or sensational topics (backlash, babies)
- visually interpretable or image-heavy content (webp, jpg, pjpg)
- technical detection or observation (detecting, delve)

These linguistic features often appear in posts that are memorable, emotionally salient, or rich in specific content, leading to higher user engagement.

When neoliberal is included, the highest-scoring words shift toward political, conflict-oriented, or socially charged terms: “shitposters,” “blamed,” “dire,” “dropping,” “protectionist,” “finances,” “rigged.” This reflects the stronger political and opinion-driven environment of the neoliberal subreddit, where high-engagement posts frequently involve debate, criticism, or satire. Even technical tokens such as “1024” and “webp” maintain positive associations, indicating that visually oriented or meme-adjacent content also performs well.



### 5.1.5 Business Insight

The analysis reveals that comments receiving higher scores tend to contain stronger elements of narrative depth, emotional expression, visual or sensory cues, and concrete informational content, whereas comments with lower scores rely more heavily on conversational fillers, formalities, or administrative language. When the neoliberal subreddit is included, this pattern becomes even more pronounced: politically charged, argumentative, or emotionally provocative terms show a substantially stronger positive influence on engagement. This indicates that in politically oriented communities, emotional and socio-political framing plays a critical role in attracting user attention and driving upvotes.

These findings offer practical implications for content creators, AI communication teams, and researchers seeking to optimize engagement in online discussions. Low-information or routine conversational expressions such as “thanks” or “yeah”, do not contribute meaningfully to visibility or user response. In contrast, comments that incorporate narrative elements, emotional resonance, or visually evocative language tend to perform significantly better. Moreover, the optimal content strategy depends on the nature of the community: technical subreddits reward analytical and information-rich language, whereas political or socially oriented communities respond more strongly to emotionally charged or provocative wording. Ultimately, the results demonstrate that word choice itself is a key driver of engagement, shaping how content circulates and how audiences respond within different segments of the AI discourse ecosystem.

## 5.2 Viral/non-Viral(Binary)

The second business question reframed the task as a binary classification problem, where a post was labeled as *viral* if its score exceeded a given threshold. Two versions were evaluated:

- **Threshold = 8:** score > 8 → viral
- **Threshold = 9:** score > 9 → viral

Model performance was assessed using ROC-AUC, PR-AUC, Accuracy, and F1.

### 5.2.1 Feature Engineering

The classifier used a feature pipeline similar to the regression model:

- **TF-IDF features** from tokenized, cleaned text
- **Topic distribution features** from LDA
- **Metadata:** day\_of\_week, hour, text\_length

Logistic Regression was used as the classification model. Coefficients enabled interpretation of which words increased (*positive coef*) or reduced (*negative coef*) the likelihood of virality.

The top 20 positive and negative coefficients for both *with* and *without neoliberal* conditions are shown in (Figure Z-AA).

### 5.2.2 Model Performance

For threshold 8, ROC-AUC improved from 0.6496 (without neoliberal) to 0.6745 (with neoliberal). Similarly, PR-AUC improved from 0.2097 to 0.3336, indicating the model benefits from the higher variance and stronger emotional tone present in the neoliberal subreddit.

For threshold 9, ROC-AUC and PR-AUC follow a similar pattern, again showing stronger performance when neoliberal is included, though accuracy and F1 are higher without neoliberal due to class imbalance effects. These results imply that text-based emotional and political language captured from broader communities helps the model detect virality signals, but may reduce accuracy by shifting class boundaries.

### 5.2.3 Words Associated With Lower Viral Probability

Across both models (with and without the neoliberal subreddit), the strongest negative coefficients correspond to conversational fillers, greetings, administrative actions, and meta-discussion terms. Words such as “thanks,” “thx,” “thank,” “dm,” “hi,” “replies,” “contact,” “removed,” and “addressed” consistently decrease the likelihood that a post will go viral.

These terms typically signal replies, thread maintenance, moderation actions, or routine interpersonal exchanges, all of which tend to appear in low-visibility comment chains rather than high-impact standalone posts. When neoliberal is included, politically adjacent administrative tokens such as “participate,” “linked,” “2fneoliberal,” “1024,” and “vpn” also emerge as strongly negative predictors, again indicating that logistical coordination or meta-thread interactions—not substantive content—fail to generate viral behavior.

### 5.2.4 Words Associated With Higher Viral Probability

Words that increase viral probability differ more sharply depending on whether the *neoliberal* subreddit is included.

#### Without neoliberal

The strongest positive predictors include terms such as:

- **“linus,” “execs,” “emad,” “giphy,” “butter,” “bell,” “hallucinated,” “demonstration,” “jpg,” “tweets,” “hilarious,” “reusable,” “plots,” “pixart,” “shocking”**

These words cluster into three clear semantic categories:

1. **Tech personalities or companies** (linus, execs, emad)
2. **Humor, memes, or visual media** (giphy, hilarious, shocking, pixart)
3. **Model behavior or failure modes** (hallucinated, demonstration, reusable)

This suggests that high-engagement content often involves recognizable public figures, emotionally expressive or humorous framing, and technical novelty or surprising model outputs.

With neoliberal

The strongest predictors become overtly political or ideological:

- “ping,” “watercooler,” “shitposters,” “pjpg,” “1170,” “zionists,” “homeowners,” “leftists,” “yimby,” “benjamin,” “headline,” “fetterman,” “subsidizing,” “unhinged,” “article”

These terms reflect:

1. **Political actors, factions, and policy topics**
2. **Debate-oriented or emotionally polarized discourse**
3. **Humor, satire, and shitposting culture**
4. **Factual claims embedded in article-like content**

### 5.2.5 Business Insight

Viral AI-related content follows a dual behavioral pattern across communities: in general AI subreddits, virality is driven by humor, visual or meme-like elements, narrative hooks, and surprising model behaviors, while in politically adjacent communities, emotionally escalated language, ideological identity markers, and conflict-framed discourse dominate as the main engines of viral spread. Across all models, low-engagement posts consistently contain conversational fillers—such as greetings, acknowledgments, and moderation or administrative language—which contribute little information and fail to attract broader attention.

These patterns offer clear strategic implications for creators and organizations seeking to maximize reach. To increase viral probability, one should minimize low-information conversational phrases (“thanks,” “hi,” “dm,” “replies”) and instead incorporate narrative, visual, or emotionally evocative elements that naturally capture user interest. Messaging should adapt to community personality: technical AI communities reward novelty, demonstrations, and model behavior, whereas political or socially oriented communities respond more strongly to emotionally charged or identity-driven content. Ultimately, the findings show that virality in AI discussions is shaped less by timing or length and more by the semantic and emotional fingerprint of the words themselves.

## 6. Discussion

### 6.1 Limitations

Despite the breadth and depth of this study, several limitations shape the interpretation of our findings.

First, dataset imbalance remains a core constraint. AskReddit alone accounts for over 58 million rows—far exceeding the size of smaller, specialized AI communities. Although stratified sampling and normalization were applied throughout the analysis, this imbalance may still amplify general-audience sentiment relative to expert discourse.

Second, sentiment and emotion classification in AI-related text introduces significant complexity. Reddit discussions frequently contain sarcasm, memes, jargon, and evolving internet language that standard lexicon-based approaches struggle to capture. Additionally, emojis and multimodal references (screenshots, code blocks, images) carry emotional content that is not directly reflected in the textual pipeline. While domain-specific cleaning and validation improved accuracy, the findings must be interpreted with an understanding that subtle affective cues may be underdetected.

Third, deleted, edited, and removed posts introduce unavoidable blind spots in subreddit ecosystems. Power users also exhibit bursty participation patterns, meaning data snapshots may miss important contextual history. Finally, differences in moderation practices across subreddits produce structural biases that shape what kinds of discourse remain visible.

Overall, these limitations highlight the need for more nuanced sentiment models and richer cross-platform datasets to fully capture the diversity of AI discourse.

### 6.2 Challenges from Dataset Size and Computation Limits

Working with 71 million rows (and more than 8 billion rows in the unfiltered corpus) required extensive engineering and imposed significant computational trade-offs.

Managing compatibility issues between Spark 3.7 and Spark 4.0, handling EC2's four-hour timeouts, and coordinating memory allocation for NLP pipelines were persistent challenges. Many analytical tasks—especially regression modeling—required hours of cluster time and careful segmentation of jobs to prevent executor failures.

Budget constraints also forced the team to offload portions of analysis to local machines, demonstrating both flexibility and the limitations of student-scale cloud resources. NLP tasks such as tokenization, TF-IDF, and LDA had to be subsetted repeatedly to remain within memory constraints. While this approach enabled the project to succeed, it also limited the extent of hyperparameter tuning and model complexity.

These computational realities underscore the difficulty of conducting large-scale social media analytics without sustained cloud resources, and they highlight the critical role of efficient pipeline design, data reduction strategies, and distributed systems literacy.

## 7. Conclusions and Next Steps

### 7.1 So what? - Key Takeaways and Implications

The overarching insight from this project is that AI discourse during the 2023–2024 boom did not converge into a single narrative. Instead, Reddit exhibited a fragmented ecosystem of parallel conversations that reflected each community's core identity:

- Developers focused on implementation barriers, reliability concerns, and optimization.
- Artists and creators discussed model instability, ethics, aesthetics, and ownership.
- Futurists and ideological communities expressed economic, existential, and societal anxieties.
- General users (AskReddit) showed the strongest emotional responses, particularly fear and sadness.
- Model-specific communities (ChatGPT, GPT-4, Claude) communicated in a stable, neutral, utilitarian tone.

From a business and research perspective, these findings have deep implications:

#### 1. Timing, not volume, drives engagement.

The best-performing posts occur during moderate-competition windows, not peak hours, meaning creators can optimize reach without posting more.

#### 2. Reliability is the real friction point.

User frustration overwhelmingly centers on inconsistency and errors—not lack of capability. This shifts product strategy away from “new features first” toward durability, stability, and trust.

#### 3. Word choice drives virality.

High-engagement posts contain narrative, emotional, or surprising language—while low-engagement posts rely on administrative or conversational filler.

#### 4. AI adoption is socially stratified.

Communities process AI differently depending on their role, values, and expectations, meaning AI companies must segment communication strategies rather than rely on universal messaging.

Ultimately, this project shows that understanding public perception of AI requires examining real conversations, not projections or surveys. Reddit's organic discourse reveals what people actually struggle with, celebrate, fear, and hope for.

## 7.2 Next Step

Building on the foundation of this project, several high-impact future directions emerge across six key areas: cross-platform comparative analysis, advanced NLP with transformer models, bot-human differentiation, predictive modeling of discourse evolution, richer engagement modeling, and longitudinal tracking of AI fatigue and adoption patterns. The following points outline each of these directions in more detail:

### 1. Cross-Platform Comparative Analysis

Expand the dataset to include Twitter/X, YouTube comments, Hacker News, Discord servers, and GitHub discussions. This will reveal how platform design and community structures shape AI narratives differently from Reddit.

### 2. Advanced NLP With Transformer Models

Fine-tune BERT, RoBERTa, or domain-adapted LLMs to better detect sarcasm, memes, technical jargon, and emoji-level sentiment. This would significantly improve emotional and semantic accuracy over lexicon-based approaches.

### 3. Bot vs Human User Detection

Identify automated accounts and analyze how bots influence discourse dynamics, misinformation, and hype cycles. Understanding bot amplification will be crucial as AI-generated content continues to increase.

### 4. Predictive Modeling of Discourse Evolution

Use sequence models or time series forecasting to predict future sentiment spikes, topic shifts, or hype cycles around major AI releases—valuable for product planning and market intelligence.

### 5. Richer Engagement Modeling

Incorporate Reddit network dynamics, thread structure, and temporal decay functions to refine virality prediction beyond simple binary thresholds.

### 6. Longitudinal Tracking of AI Fatigue and Adoption Patterns

As generative AI becomes normalized, studying how enthusiasm, frustration, and expectations evolve will offer insight into long-term adoption readiness across different user groups.